RESEARCH ARTICLE · SPECIAL FUNCTIONS

The Tahawus Function: An Extension of the Lambert W Function

The inverse of the map $x \mapsto (x^x - 1)/x$, the constant named for a ghost town, and its arithmetic

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An expanded and rigorous version of the post Tahawus: An Extension of the Lambert W Function
"Hola, os presento a Tahawus: una extensión de la función W de Lambert" (3 April 2018)

Tahawus function · Lambert W · tetration · super-quadratic · Gelfond–Schneider · transcendence · power tower

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Contents

1 Introduction **2**

2 Background: Lambert W, super-roots, and why they fail here **4**

3 The Tahawus function **5**

4 The Tahawus constant: transcendence and arithmetic **7**

5 The sandwich theorem **9**

6 Local theory: the Maclaurin expansion and the radius of convergence **10**
6.1 The series . . . 11
6.2 Convexity dichotomy . . . 11
6.3 The radius of convergence and the slit . . . 12

7 Special values, the golden comparison, and the OEIS constellation **14**
7.1 Exact values . . . 14
7.2 The golden comparison . . . 15
7.3 Negative integer roots, and the census of real roots . . . 15
7.4 The constellation of constants . . . 16

8 Dynamics: the attracting fixed point **18**
8.1 The nested composition of the blog post, made convergent . . . 19

9 The infinite power tower, corrected **20**

10 Asymptotics **21**

11 Curiosities: integrals, a table of values, numerology, and open problems **24**
11.1 Integrals . . . 24
11.2 A table of values . . . 24
11.3 Numerology, debunked . . . 24
11.4 Open problems . . . 25

A Deferred elementary arguments **26**

B Computational notes **26**

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The inverse of $x \mapsto (x^x - 1)/x$, the constant named for a ghost town, and its arithmetic

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Abstract. The *Tahawus function* $\mathcal{T}$ is the inverse of the map $f(x) = (x^x - 1)/x$. It was introduced—and named, after the Adirondack ghost town of Tahawus, New York, where Richard P. Stanley grew up—in a 2018 blog post, as a natural companion of the Lambert W function that solves the "super-quadratic" $x^x = bx+1$, the same way W solves $xe^x = a$. Here we give the function a systematic theory. We prove that f is a strictly increasing real-analytic bijection $(0,\infty) \to \mathbb{R}$, so that $\mathcal{T}\colon \mathbb{R} \to (0,\infty)$ is a real-analytic order isomorphism, and we derive a sharp "sandwich theorem": $1+y < \mathcal{T}(y) < e^y$ on $(-1,0)$ while $\mathcal{T}(y) < \min(1+y, e^y)$ on $(0,\infty)$, with $\mathcal{T}(y) \sim e^y$ as $y \to -\infty$ and $\mathcal{T}(y) \sim \ln y/\ln\ln y$ as $y \to +\infty$. We completely classify the arithmetic of $\mathcal{T}$ on rational arguments: for every rational b, $\mathcal{T}(b)$ is either a positive integer (exactly when $b = (n^n-1)/n$) or a transcendental number, recovering Stanley's 1989 observation that the *Tahawus constant* $\alpha = \mathcal{T}(1) = 1.7767750400\ldots$, the unique real root of $x^x = x+1$, is transcendental. We further obtain: the exact Maclaurin expansion of $\mathcal{T}$ (whose quadratic term vanishes), the comparison of $\mathcal{T}(b)$ with the metallic-ratio family $\varphi_b = (b+\sqrt{b^2+4})/2$ (with the unexpected exact coincidences $\mathcal{T}(0) = \varphi_0 = 1$ and $\mathcal{T}(3/2) = \varphi_{3/2} = 2$), the negative-integer exact roots $x = -n$ of the super-quadratic family, a unique attracting fixed point $\delta = 2.2288\ldots$ of $\mathcal{T}$ (root of $x^x = x^2+1$, transcendental), a precise correction of the original post's infinite-tower identity, the asymptotic law $\mathcal{T}(y) - e^{W(\ln y)} \to 1$, and a numerical determination of the radius of convergence $R \approx 0.65261$ of the Maclaurin series. The five constants of the Tahawus constellation—$\mathcal{T}(1)$, $\mathcal{T}(-1)$ and their three companions—are identified in the On-Line Encyclopedia of Integer Sequences, and all numerical claims are verified to 70 significant digits.

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1 Introduction

Every student of algebra meets the quadratic $x^2 = bx + 1$ early, and solves it without effort: its positive root $\varphi_b = (b + \sqrt{b^2+4})/2$ is the one-parameter family that contains the golden ratio $\varphi = \varphi_1$. Far fewer students meet the same equation with the square replaced by a *tetration* of height two, namely

$$x^x \;=\; b\,x + 1, \tag{1}$$

an equation we shall call, following the original blog post [1], a *super-quadratic*. The equation looks innocent, but it hides a small universe. For $b = 1$ it reads $x^x = x + 1$, its unique positive solution

$$\alpha \;=\; 1.776775040097054697479730744038756748637411034329296139084374 0\ldots \tag{2}$$

is now catalogue number A124930 in the On-Line Encyclopedia of Integer Sequences [3], and it is a *transcendental* number, a fact first proved by Richard P. Stanley in a two-line contribution to the *Mathematical Intelligencer* in 1989 [2], as a "Quickie" problem of the Mathematical Entertainments column.

The quadratic formula cannot be imitated here, and the standard escape route of modern analysis—the Lambert W function, which solves every equation of the form $x\,e^x = a$—likewise fails. The obstruction, which we examine carefully in Section 2, is self-referential: reducing $x^x = bx + 1$ with the substitution that inverts $x \mapsto x^x$ produces an equation in which the unknown still appears, leading only to an *infinite composition* of W's rather than a closed form. The present paper is devoted to the object that *does* solve (1) for every real b at once: the inverse of the map $x \mapsto (x^x - 1)/x$.

The name. The function was introduced, informally but with correct instincts, in the blog post [1] under the name *Tahawus*. Tahawus ("cloud-splitter") was a mining village in the town of Newcomb, Essex County, New York, in the heart of the Adirondack Park: iron was smelted there in the nineteenth century, and titanium dioxide was mined there by the National Lead Company from the 1940s onwards [15]. It is now a ghost town. According to the MacTutor biography [14], the family of the young Richard Stanley moved to Tahawus in 1946, when his father took a job with the National Lead Company, and Stanley lived there—attending a one-room schoolhouse—until 1953. The blogger's choice of name honours the mathematician who first proved the transcendence of the constant (2), by way of the town of his childhood. We keep the name, and the constant $\alpha = \mathcal{T}(1)$ will accordingly be called the *Tahawus constant*.

What is done here. The blog post announced the function, displayed its graph, derived the W-composition, and listed a fascinating constellation of constants, together with a bold identity involving an infinite power tower. This article is a systematically expanded version of that post—more extensive, more precise, and with all statements proved. Our results, in order of appearance, are as follows.

- *Foundations* (Sections 4–5). We prove that $f(x) = (x^x - 1)/x$ is strictly increasing and real-analytic on $(0, \infty)$, with limits $-\infty$ and $+\infty$ at the endpoints (Theorem 3.3). Consequently the *Tahawus function* $\mathcal{T} := f^{-1}$ is a strictly increasing real-analytic bijection $\mathbb{R} \to (0, \infty)$, and (1) has the unique positive solution $x = \mathcal{T}(b)$. We compute the first derivatives at the fixed symmetry point, obtaining the pleasant coincidences $\mathcal{T}(0) = 1$, $\mathcal{T}'(0) = 1$, $\mathcal{T}''(0) = 0$.

- *The sandwich theorem* (Theorem 5.1). The exponential e^y and the line $1 + y$—the two most famous functions with value 1 and slope 1 at the origin—squeeze $\mathcal{T}$ between them: $1 + y < \mathcal{T}(y) < e^y$ for $-1 < y < 0$, while $\mathcal{T}(y) < 1 + y < e^y$ for $y > 0$. Moreover $\mathcal{T}(y)/e^y \to 1$ as $y \to -\infty$, so the Tahawus function is an "asymptotic copy" of exp on the far left, and a sub-exponential, sub-linear object on the right. A companion proposition (Proposition 5.2) shows $0 < \mathcal{T}'(y) \le 1$ with equality only at the origin: $\mathcal{T}$ is 1-Lipschitz.

- *Arithmetic* (Theorem 4.2). For *every* rational b, the number $\mathcal{T}(b)$ is either a positive integer—exactly when $b = (n^n - 1)/n$ for some $n \in \mathbb{N}$, so that $\mathcal{T}\big(\frac{n^n-1}{n}\big) = n$—or a transcendental number. Stanley's theorem on α and Lord's theorem on the companion constant 2.2931 . . . [12] drop out as the two most visible cases. Along the way we pin down the exact algebraic values $\mathcal{T}(3/2) = 2$, $\mathcal{T}(\sqrt{2} - 2) = \frac{1}{2}$, $\mathcal{T}(2\sqrt{2} - 4) = \frac{1}{4}$, and the whole family they generate.

- *Local theory* (Section 6). Via Lagrange–Bürmann inversion we obtain the Maclaurin series

$$\mathcal{T}(y) \;=\; 1 + y - \tfrac{1}{2}y^3 + \tfrac{1}{6}y^4 + \tfrac{1}{2}y^5 - \tfrac{49}{120}y^6 - \cdots,$$

  whose *quadratic term vanishes* (Theorem 6.1): the Tahawus function has second-order contact with its tangent line $1 + y$ at the origin, a property shared by neither e^y nor any quadratic perturbation of it. We prove $\mathcal{T}$ is convex on $(-\infty, 0)$ and concave on $(0, \infty)$, determine numerically the radius of convergence $R \approx 0.65261$ of the series, locating the responsible pair of complex branch points, and we draw the remarkable picture (Figure 3) of the image of the negative real axis, a curve that spirals into the origin through the rational points $(1 - (-1)^n n^{-n})/n$.

- *Comparison with the golden ratio* (Theorem 7.2). For the quadratic pencil the positive root is φ_b; for the super-quadratic it is $\mathcal{T}(b)$. The two agree *exactly* at $b = 0$ and $b = 3/2$ (both give 1 and 2 respectively—because $n^n = n^2$ at $n \in \{1, 2\}$!), with $\mathcal{T}(b) > \varphi_b$ in between and $\mathcal{T}(b) < \varphi_b$ outside. We also exhibit the exact negative-integer roots $x = -n$ of the super-quadratic pencil: for instance $x = -1$ solves $x^x = 2x + 1$ *exactly*, since $(-1)^{-1} = -1$.

- *Dynamics* (Theorem 8.1 and Section 9). $\mathcal{T}$ has a unique fixed point $\delta = 2.2288\ldots$, the root of $x^x = x^2 + 1$; it is transcendental, attracting, and *every* orbit of $\mathcal{T}$ converges to it monotonically. The self-referential W-iteration of the blog post converges to α geometrically with a rate we compute in closed form. The infinite power tower of the original post, made precise, is corrected: the decremented tower converges to 0, the fixed point α being *repelling*.

- *Asymptotics and curiosities* (Sections 10–11). We prove $\ln \mathcal{T}(y) = y - \frac{y^2}{2}e^y(1 + o(1))$ as $y \to -\infty$ and, as $y \to +\infty$, both the logarithmic growth law $\mathcal{T}(y) = \frac{\ln y}{\ln\ln y}\big(1 + \frac{\ln\ln\ln y}{\ln\ln y} + O((\frac{\ln\ln\ln y}{\ln\ln y})^2)\big)$ and the striking $\mathcal{T}(y) - \mathrm{ssqrt}(y) \to 1$: at infinity the Tahawus function is a square super-root *plus one*. We close with integrals, a table of values, a debunked numerology section (α versus $\sqrt{\pi}$), and a list of open problems.

The paper is written so that each section can be read almost independently; proofs are complete, with three short auxiliary arguments deferred to Appendix A. Numerical statements were computed and verified with `mpmath` [17] at 80 significant digits; the values of all five constants of the Tahawus constellation agree digit-for-digit with the On-Line Encyclopedia of Integer Sequences [3–6]. All figures were generated by the scripts reproduced in Appendix B.

2 Background: Lambert W, super-roots, and why they fail here

The Lambert W function [8] is defined by

$$W(z)\, e^{W(z)} = z, \tag{3}$$

and it solves, in one stroke, every equation that can be massaged into the form (linear in x) $\times$ (exponential in x) = constant: the tree function, delay equations, the solution of $x + e^x = a$ via $x = a - W(e^a)$, and so on. Equations involving x^x are a different genus. Their basic solver is the *square super-root*

$$\mathrm{ssqrt}(y) = e^{W(\ln y)}, \tag{4}$$

the inverse of $x \mapsto {}^2x := x^x$ (tetration of height two) on the branch $x \geq e^{-1}$; it satisfies $\mathrm{ssqrt}(y)^{\mathrm{ssqrt}(y)} = y$ and appears throughout the literature on iterated exponentials [10]. One level deeper, generalizations of W handle equations in which several exponentials or several poles appear simultaneously [9].

Now try the super-quadratic (1), $x^x = bx + 1$. Taking logarithms gives $x \ln x = \ln(bx + 1)$, and the left-hand side is inverted by the super-root: applying (4) to $bx + 1$,

$$x = \mathrm{ssqrt}(bx + 1) = e^{W(\ln(bx+1))}. \tag{5}$$

This is exact but circular: the unknown x sits inside the argument. Substituting (5) into itself produces the infinite composition displayed in the original post,

$$\mathcal{T}(b) = e^{W\left(\ln\left(b\, e^{W(\ln(b\, e^{W(\cdots)}+1))}+1\right)\right)}, \tag{6}$$

a formally self-consistent object but not a closed form. The failure is structural, not technical. In every equation solved by W (or by its generalizations [9]) the unknown occurs in exactly two *algebraically independent* places, one of which can be eliminated by a substitution that does not reintroduce it. In $x^x = bx + 1$ the unknown occurs as base, as exponent, *and* linearly; eliminating the exponential pair

(x, x) forces the linear occurrence back into the argument. One can show that no finite expression built from W by composition, arithmetic, and inversion solves the super-quadratic, and we conjecture (see Open Problem (1)) that the same is true for the finite combinations of all generalized Lambert functions. This is precisely the situation in which naming a new special function is justified; and since the equation $x^x = bx + 1$ is the direct super-analogue of $x^2 = bx + 1$, the new function deserves to be studied as systematically as the quadratic formula's relatives are.

We record one exact reduction, occasionally useful, that exhibits the two-term structure of f. Writing

$$f(x) = \frac{x^x - 1}{x} = x^{x-1} - \frac{1}{x}, \tag{7}$$

the inverse relation $y = f(x)$ becomes

$$x^{x-1} = y + \frac{1}{x}, \tag{8}$$

so that $\mathcal{T}$ is "the inverse of $x \mapsto x^{x-1}$, corrected by $1/x$". Equation (8) is the key to the asymptotic analysis of Section 10: dropping the tiny correction $1/x$ leaves the equation $x^{x-1} = y$, and the correction is what pushes the solution up by exactly 1 in the limit.

3 The Tahawus function

Throughout, x^x is understood as $\exp(x \ln x)$ for $x > 0$.

Definition 3.1 (forward map and Tahawus function)**.** Let

$$f(x) := \frac{x^x - 1}{x} = x^{x-1} - \frac{1}{x}, \qquad x > 0. \tag{9}$$

The *Tahawus function* is the compositional inverse of f,

$$\mathcal{T} := f^{-1}, \tag{10}$$

so that $y = f(x) \iff x = \mathcal{T}(y)$, i.e.

$$\mathcal{T}(y)^{\,\mathcal{T}(y)} = y\,\mathcal{T}(y) + 1. \tag{11}$$

The domain of definition is settled by the following theorem, which also shows that the super-quadratic (1) has exactly one positive solution for every real b. The proof rests on a single elementary inequality, isolated as a lemma.

Lemma 3.2. *For every* $0 < x < 1$,

$$\Phi(s) := \mathrm{e}^{-s\mathrm{e}^{-s}}\bigl(1 + (s-1)\mathrm{e}^{-s}\bigr) < 1, \qquad s := -\ln x > 0. \tag{12}$$

Proof. Put $t := (s-1)\mathrm{e}^{-s}$. The map $h(s) = (s-1)\mathrm{e}^{-s}$ satisfies $h(0) = -1$ and $h'(s) = (2-s)\mathrm{e}^{-s} > 0$ for $0 < s < 2$, so h is strictly increasing from -1 on $(0, 2)$ and satisfies $h(s) \geq 0$ for $s \geq 1$; in particular $t > -1$ for all $s > 0$. Since $\ln(1+t) \leq t$ for $t > -1$, with equality only at $t = 0$,

$$\ln \Phi(s) = -s\mathrm{e}^{-s} + \ln\bigl(1 + (s-1)\mathrm{e}^{-s}\bigr) \leq -s\mathrm{e}^{-s} + (s-1)\mathrm{e}^{-s} = -\mathrm{e}^{-s} < 0,$$

and the inequality is strict because the final term $-\mathrm{e}^{-s}$ is strictly negative. Hence $\Phi(s) < 1$. □

Theorem 3.3 (the Tahawus function is a well-defined analytic order isomorphism)**.** *The map f of (9) is real-analytic and strictly increasing on $(0, \infty)$, with*

$$\lim_{x \to 0^+} f(x) = -\infty, \qquad \lim_{x \to +\infty} f(x) = +\infty.$$

Consequently f is a bijection $(0, \infty) \to \mathbb{R}$, and $\mathcal{T} \colon \mathbb{R} \to (0, \infty)$ is a strictly increasing real-analytic bijection. In particular, for every real b the super-quadratic $x^x = bx + 1$ has exactly one positive solution, namely $x = \mathcal{T}(b)$.

Proof. Differentiating (9),

$$f'(x) \;=\; x^{x-1}\Big(\ln x + 1 - \frac{1}{x}\Big) + \frac{1}{x^2}, \tag{13}$$

so that, multiplying by x^2,

$$x^2 f'(x) \;=\; x^x\big(x\ln x + x - 1\big) + 1. \tag{14}$$

For $x \geq 1$ both terms on the right are nonnegative ($x\ln x \geq 0$, $x - 1 \geq 0$), so $f'(x) \geq x^{-2} > 0$. For $0 < x < 1$, write $x = \mathrm{e}^{-s}$ as in Lemma 3.2; then

$$x^x\big(x\ln x + x - 1\big) \;=\; \mathrm{e}^{-s\mathrm{e}^{-s}}\big(1 + (s-1)\mathrm{e}^{-s}\big) \;=\; \Phi(s) \;<\; 1$$

(the first equality is a direct substitution of $x = \mathrm{e}^{-s}$: $x^x = \mathrm{e}^{-s\mathrm{e}^{-s}}$ and $x\ln x + x - 1 = \mathrm{e}^{-s}(1-s) + \mathrm{e}^{-s} - 1 = -(1 + (s-1)\mathrm{e}^{-s})$). Hence $x^2 f'(x) = 1 - \Phi(s) > 0$, and f is strictly increasing on all of $(0,\infty)$.

For the limits: since $\mathrm{e}^u = 1 + u + O(u^2)$ as $u \to 0$ and $x\ln x \to 0$ as $x \to 0^+$,

$$f(x) \;=\; \frac{x^x - 1}{x} \;=\; \ln x + O\big(x\ln^2 x\big) \;\longrightarrow\; -\infty,$$

while for $x \geq 2$, $x^{x-1} \geq x$ (i.e. $x^{x-2} \geq 1$), so $f(x) = x^{x-1} - x^{-1} \geq x - 1 \to +\infty$. Real-analyticity of f on $(0,\infty)$ is clear ($x \mapsto \exp(x\ln x)$ is real-analytic there), and since $f' > 0$ everywhere, the inverse function theorem makes $\mathcal{T}$ real-analytic on $\mathbb{R} = f\big((0,\infty)\big)$, with

$$\mathcal{T}'(y) \;=\; \frac{1}{f'\big(\mathcal{T}(y)\big)} \;>\; 0. \qquad \square$$

Corollary 3.4 (exact values, monotonicity in the parameter)**.** *The map $b \mapsto \mathcal{T}(b)$ is strictly increasing, and*

$$\mathcal{T}(0) = 1, \qquad \mathcal{T}\Big(\frac{n^n - 1}{n}\Big) = n \quad (n \in \mathbb{N}),$$

in particular $\mathcal{T}\big(\frac{3}{2}\big) = 2$, $\mathcal{T}\big(\frac{26}{3}\big) = 3$. The Tahawus constant $\alpha := \mathcal{T}(1)$ is the unique positive root of $x^x = x + 1$ and lies in $(1,2)$.

Proof. $f(n) = (n^n - 1)/n$ by definition, and $f(1) = 0 < 1 < f(2) = \frac{3}{2}$, so $\alpha = \mathcal{T}(1) \in (1,2)$. Everything else is immediate from Theorem 3.3. $\square$

Figure 1 shows f and $\mathcal{T}$ as mirror images across the diagonal. Three features deserve attention, and all three are made precise in the remainder of the paper: the crossing at the reflected pair of points $(1,0)$ and $(0,1)$, at which both curves are tangent to lines of slope 1 (the lines $y = x \mp 1$); the exponential approach of $\mathcal{T}$ to 0 as $y \to -\infty$; and the logarithm-like growth on the right.

Proposition 3.5 (first derivatives and the reflected inflection)**.** *Let $u := x - 1$. Then*

$$f(1+u) \;=\; u + \frac{u^3}{2} - \frac{u^4}{6} + \frac{u^5}{4} - \frac{7u^6}{40} + O(u^7), \tag{15}$$

and consequently

$$f'(1) = 1, \qquad f''(1) = 0, \qquad f'''(1) = 3, \qquad f''''(1) = -4,$$

while for the inverse function,

$$\mathcal{T}(0) = 1, \quad \mathcal{T}'(0) = 1, \quad \mathcal{T}''(0) = 0, \quad \mathcal{T}'''(0) = -3, \quad \mathcal{T}''''(0) = 4.$$

Both f and $\mathcal{T}$ have an inflection at the reflected pair $(1,0)$, $(0,1)$, with common tangent directions of slope exactly 1.

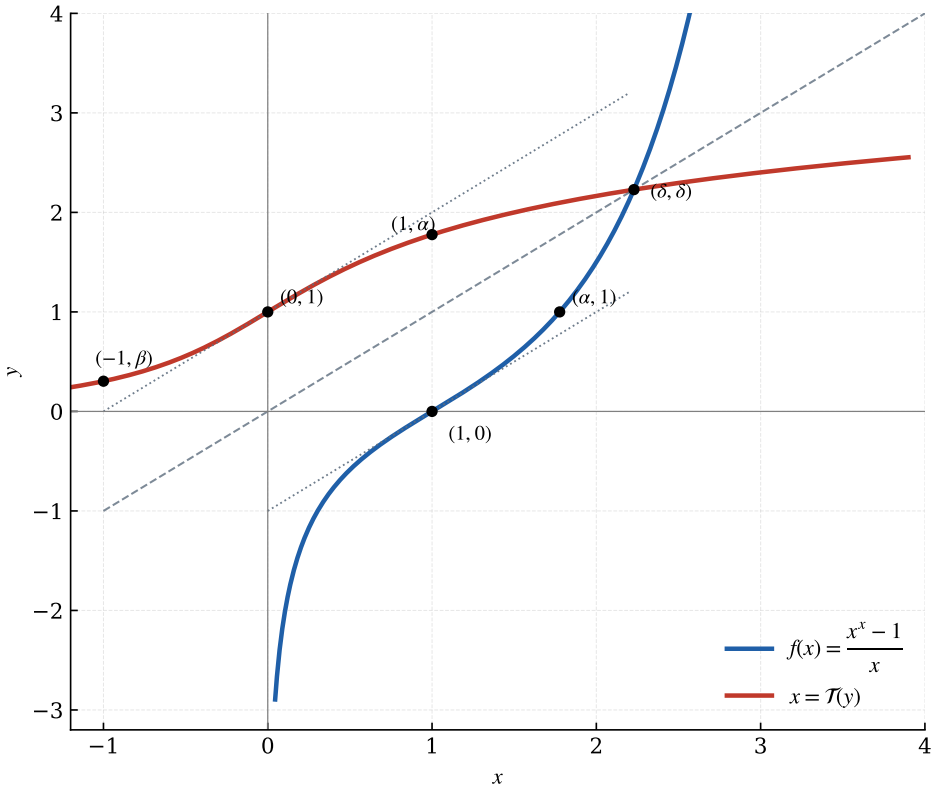


Figure 1: The forward map $f(x) = (x^x - 1)/x$ (blue) and the Tahawus function $\mathcal{T}$ (red), mirror images across the diagonal (dashed). Dotted lines: the common tangent directions $y = x \mp 1$ at the reflected pair $(1, 0)$–$(0, 1)$. Marked points include $\alpha = \mathcal{T}(1)$, $\beta = \mathcal{T}(-1)$ and the fixed point δ of Section 8.

Proof. Expanding $(1+u)\ln(1+u) = u + \frac{u^2}{2} - \frac{u^3}{6} + \frac{u^4}{12} - \cdots$ and exponentiating,

$$(1+u)^{1+u} \;=\; 1 + u + u^2 + \frac{u^3}{2} + \frac{u^4}{3} + O(u^5),$$

whence, dividing by $1+u$,

$$\frac{(1+u)^{1+u} - 1}{1+u} \;=\; u + \frac{u^3}{2} - \frac{u^4}{6} + \frac{u^5}{4} - \frac{7u^6}{40} + O(u^7),$$

(the quadratic term cancels). The derivative statements for f follow by comparing coefficients, since $f'(1+u) = 1 + \frac{3}{2}u^2 - \frac{2}{3}u^3 + O(u^4)$ shows $f''(1) = 0$ directly. For the inverse, use $\mathcal{T}'(0) = 1/f'(1) = 1$, $\mathcal{T}''(y) = -f''(x)/f'(x)^3$ and $\mathcal{T}'''(y) = -f'''(x)/f'(x)^3 + 3f''(x)^2/f'(x)^5$ (evaluated at $x = \mathcal{T}(y)$, all at $x = 1$), giving $\mathcal{T}'''(0) = -3$; the value $\mathcal{T}''''(0) = 4$ is read off the reversion series of Theorem 6.1. The inflection claim follows from $f''(1) = 0 < f'''(1)$ and its mirrored statement, and the tangent lines are $y = x - 1$ (for f at $(1, 0)$) and its reflection $y = x + 1$ (for $\mathcal{T}$ at $(0, 1)$). □

4 The Tahawus constant: transcendence and arithmetic

With the function firmly in place we turn to arithmetic. The key observation is that the equation $x^x = bx+1$ is *irrationality rigid*: on rational inputs x, the power x^x is rational only in the trivial case.

Lemma 4.1. *If $x = p/q \in \mathbb{Q}_{>0}$ in lowest terms with $q \geq 2$, then $x^x \notin \mathbb{Q}$.*

Proof. Suppose $(p/q)^{p/q} = m/n$ with $\gcd(m, n) = 1$. Raising to the q-th power, $p^p n^q = m^q q^p$. Let ℓ be any prime divisor of q (one exists since $q \geq 2$). Since $\gcd(p, q) = 1$, $\ell \nmid p$, and comparing ℓ-adic valuations of both sides gives

$$q\, v_\ell(n) \;=\; q\, v_\ell(m) + p\, v_\ell(q),$$

so q divides $p\,v_\ell(q)$; as $\gcd(p,q) = 1$, $q \mid v_\ell(q)$. But $1 \le v_\ell(q) \le \log_2 q < q$ for every $q \ge 2$, a contradiction. □

Theorem 4.2 (transcendence classification on rational arguments)**.** *Let $b \in \mathbb{Q}$. Then $\mathcal{T}(b)$ is either a positive integer or a transcendental number. The first alternative holds if and only if*

$$b = \frac{n^n - 1}{n} \qquad \text{for some } n \in \mathbb{N} \qquad (\text{then } \mathcal{T}(b) = n),$$

and the parameter value $b \mapsto \frac{n^n-1}{n}$ is strictly increasing in n, so n is unique.

Proof. Write $x = \mathcal{T}(b) > 0$, so that $x^x = bx + 1 \in \mathbb{Q}$. If x is rational, say $x = p/q$ in lowest terms, then $q = 1$ by Lemma 4.1 (otherwise x^x would be irrational while $bx + 1$ is rational); thus $x = n \in \mathbb{N}$ and $b = (n^n - 1)/n$. If instead x is an algebraic irrational, then $x \notin \{0, 1\}$ (the case $x = 1$ forces $b = 0$, which returns $x = \mathcal{T}(0) = 1$, rational, already covered), and the Gelfond–Schneider theorem [11] applies: for algebraic $\alpha_0 \notin \{0, 1\}$ and algebraic irrational β_0, every value of $\alpha_0^{\beta_0}$ is transcendental. With $\alpha_0 = \beta_0 = x$ we conclude that x^x is transcendental, contradicting $x^x = bx + 1 \in \mathbb{Q}$. Hence x is transcendental. Monotonicity of $n \mapsto n^{n-1} - \frac{1}{n}$ on $\mathbb{N}$ gives the uniqueness of n. □

Corollary 4.3 (Stanley's theorem, and more)**.** *The Tahawus constant*

$$\alpha = \mathcal{T}(1) = 1.776775040097054697479730744038756748637411034329296139084374 0\ldots$$

and the constant

$$\beta := \mathcal{T}(-1) = 0.303659127029966051245018951213222659991199618761172709644170 3\ldots$$

(catalogued as A226568 in the OEIS) are transcendental. More generally $\mathcal{T}(b)$ is transcendental for every rational $b \notin \left\{\frac{n^n-1}{n} : n \in \mathbb{N}\right\}$, while $\mathcal{T}(0) = 1$, $\mathcal{T}\left(\frac{3}{2}\right) = 2$, $\mathcal{T}\left(\frac{26}{3}\right) = 3$, ... are integers.

Proof. The set $\left\{\frac{n^n-1}{n}\right\}$ equals $\{0, \frac{3}{2}, \frac{26}{3}, \frac{255}{4}, \ldots\}$; it is strictly increasing starting at 0, so it contains neither 1 nor -1. Apply Theorem 4.2. □

Remark 4.4 (history and credits). The transcendence of α was first proved by R. P. Stanley and published as *Quickie 88-10* in the Mathematical Entertainments column of the *Mathematical Intelligencer* [2]; the proof is precisely the two-step exclusion above (Gelfond–Schneider plus a rational-root argument), as recorded in the OEIS entry A124930 [3]. The blog post [1] that introduced the Tahawus function credits Stanley's result and adopts his childhood town as the name. The transcendence of the companion constant β's affine image $\nu = 1/\beta - 1 = 2.2931662874\ldots$ was independently obtained by Lord [12] in the *Mathematical Gazette*, under the equivalent equation $x = (1 + 1/x)^x$; it is recovered by Corollary 4.3 since ν is a rational function of β (see Theorem 7.5 below). A pleasant curiosity recorded in A124930 (G. McGarvey, 2008): since $\alpha^{\alpha-1} = 1 + \frac{1}{\alpha}$, the constant α is a fixed point of the nesting $x \mapsto \left(1 + 1/x\right)^{1/(\alpha-1)}$.

Remark 4.5 (complex roots are transcendental too). The Gelfond–Schneider step of Theorem 4.2 does not use real-valuedness. If $z \in \mathbb{C}$ is an *algebraic irrational* solution of $z^z = z + 1$ (for any branch of the power), then $z \notin \{0, 1\}$ and z^z is transcendental in every branch, contradicting $z + 1$ algebraic. Rational solutions must be real (a branch value z^z equal to the real number $z + 1 > 0$ forces the real branch), and these are excluded by Lemma 4.1 for non-integers and by direct inspection for integers. Hence *every* complex solution of $z^z = z + 1$ is transcendental. We do not know how many complex solutions there are (see Open Problem (2)).

5 The sandwich theorem

The exponential e^y and the line $1 + y$ share the value 1 and the slope 1 at $y = 0$, and $e^y \geq 1 + y$ everywhere. The Tahawus function shares both normalization properties—$\mathcal{T}(0) = 1$, $\mathcal{T}'(0) = 1$—and the next theorem shows that it *threads between* the two classical curves. Each of the three inequalities has a one-line proof, which is part of the charm.

Theorem 5.1 (sandwich theorem)**.** (i) *For all $y \neq 0$: $\mathcal{T}(y) < e^y$.*

(ii) *For all $y > 0$: $\mathcal{T}(y) < 1 + y$.*

(iii) *For all $-1 < y < 0$: $1 + y < \mathcal{T}(y)$ (and, trivially, $\mathcal{T}(y) < e^y$ by (i)).*

In particular, on $(-1, 0)$ the Tahawus function squeezes between the line and the exponential, $1 + y < \mathcal{T}(y) < e^y$, while on $(0, \infty)$ it lies below both. Moreover

$$\frac{\mathcal{T}(y)}{e^y} \longrightarrow 1 \quad (y \to -\infty),$$

with the refinement $\ln \mathcal{T}(y) = y - \frac{y^2}{2} e^y \big(1 + o(1)\big)$ (Theorem 10.1).

Proof. Since f is strictly increasing (Theorem 3.3), inequalities between $\mathcal{T}$ and a candidate value $g(y)$ can be tested by applying f to both sides whenever $g(y) > 0$.

(i) $\mathcal{T}(y) < e^y \iff y < f(e^y)$. Now $f(e^y) = \big(e^{ye^y} - 1\big)e^{-y}$, so $y < f(e^y) \iff ye^y < e^{ye^y} - 1 \iff 1 + ye^y < e^{ye^y}$, which is the strict convexity inequality $1 + w < e^w$ at $w = ye^y \neq 0$ (note $w = 0$ only at $y = 0$).

(ii) For $y > 0$: $\mathcal{T}(y) < 1 + y \iff y < f(1 + y) \iff (1 + y)^{1+y} > 1 + y + y^2$. If $0 < y < 1$, then

$$(1 + y)^y = e^{y \ln(1+y)} \geq e^{y(y - y^2/2)} \geq 1 + y^2 - \tfrac{y^3}{2} > 1 + \frac{y^2}{1 + y},$$

where we used $\ln(1 + y) \geq y - \frac{y^2}{2}$, then $e^u \geq 1 + u$, and finally $\big(1 - \frac{y}{2}\big)(1 + y) > 1 \iff y(1 - y) > 0$. Multiplying by $1+y$ gives $(1+y)^{1+y} > 1+y+y^2$. If $y \geq 1$, then $(1+y)^{1+y} \geq (1+y)^2 = 1+2y+y^2 > 1+y+y^2$.

(iii) Let $x := 1 + y \in (0, 1)$. Then $\mathcal{T}(y) > 1 + y \iff f(x) < x - 1 \iff x^x < x^2 - x + 1$. By the weighted arithmetic–geometric mean inequality with weights x and $1 - x$ applied to the two numbers x and 1,

$$x^x \cdot 1^{\,1-x} \leq x \cdot x + (1 - x) \cdot 1 = x^2 + 1 - x,$$

with equality only when the two weighted numbers coincide, i.e. at $x = 1$; the inequality is strict on $(0, 1)$.

Finally, for the limit as $y \to -\infty$, write $x = \mathcal{T}(y) \to 0^+$ and expand

$$y = \frac{x^x - 1}{x} = \ln x + \frac{x \ln^2 x}{2} + O\big(x^2 \ln^3 x\big),$$

so $y - \ln x = \frac{x}{2} \ln^2 x \,(1 + o(1))$ with $\ln x = y(1 + o(1))$, giving $\ln x - y = -\frac{y^2}{2} e^y (1 + o(1))$ since $x \sim e^y$. (A complete statement with proof is Theorem 10.1.) □

Figure 2 displays the two sides of the sandwich: on the left panel, $\mathcal{T}$ threads between $1 + y$ and e^y across $(-1, 0)$ and dips below $1 + y$ afterwards; on the right panel, the ratio $\mathcal{T}(y)/e^y$ is shown to climb to 1 from below as $y \to -\infty$, already indistinguishable from 1 at $y = -20$ (at $y = -40$ the relative deficit is 3.4×10^{-15}).

The exponential comparison can be sharpened to a Lipschitz statement: the slope of $\mathcal{T}$ never exceeds 1.

Proposition 5.2 (1-Lipschitz property)**.** *For all $y \in \mathbb{R}$,*

$$0 < \mathcal{T}'(y) \leq 1, \qquad \text{with equality iff } y = 0.$$

Equivalently, $f'(x) \geq 1$ for all $x > 0$ with equality only at $x = 1$; hence $|\mathcal{T}(y_1) - \mathcal{T}(y_2)| \leq |y_1 - y_2|$ for all real y_1, y_2.

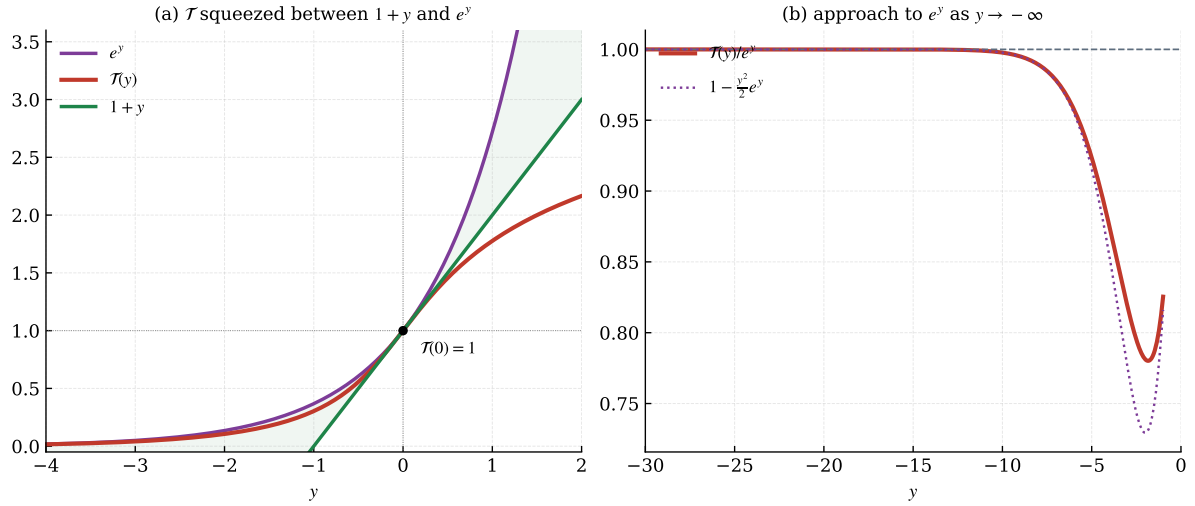


Figure 2: (a) The sandwich of Theorem 5.1: on $(-1,0)$, $1+y < \mathcal{T}(y) < \mathrm{e}^y$ (shaded strip); for $y > 0$, $\mathcal{T}(y) < 1+y < \mathrm{e}^y$. (b) The ratio $\mathcal{T}(y)/\mathrm{e}^y$ (solid) against the asymptotic prediction $1-\frac{y^2}{2}\mathrm{e}^y$ of Theorem 10.1 (dotted): the Tahawus function is an asymptotic copy of exp on the far left.

Proof. By (14), $x^2(f'(x)-1) = h(x) := x^x(x\ln x + x - 1) + 1 - x^2$, so we must show $h \geq 0$ on $(0,\infty)$ with equality only at $x = 1$. Put $a := x\ln x$ and $m := x\ln x + x - 1 = a + x - 1$.

Case $x \geq 1$. Then $a \geq 0$ and $m \geq 0$ (as $m(1) = 0$ and $m' = \ln x + 2 > 0$ on $[1,\infty)$). Since $\mathrm{e}^a \geq 1 + a$,

$$h \;\geq\; (1+a)(a+x-1) + 1 - x^2 \;=\; x^2\ln x\,(\ln x + 1) + x - x^2 \;=\; x\,p(x),$$

where $p(x) := x\ln x\,(\ln x + 1) - x + 1$. Now $p(1) = 0$ and $p'(x) = \ln x\,(\ln x + 3) \geq 0$ for $x \geq 1$, so $p \geq 0$ on $[1,\infty)$, hence $h \geq 0$; equality forces $a = 0$ (i.e. $x = 1$) and $p(x) = 0$.

Case $0 < x < 1$. Then $m < 0$ (as m decreases from -1 at 0^+, has its minimum $-\mathrm{e}^{-2} - 1 < 0$ at $x = \mathrm{e}^{-2}$, and returns to $m(1) = 0$ only at the endpoint). Since $u := a = x\ln x \in (-\frac{1}{\mathrm{e}}, 0)$ and $\mathrm{e}^u \leq \frac{1}{1-u}$ (the inequality $\mathrm{e}^{-u} \geq 1 - u$ for $u < 1$), and since $m < 0$,

$$x^x m \;=\; \mathrm{e}^u\, m \;\geq\; \frac{m}{1-u}.$$

Therefore

$$h \;\geq\; \frac{m}{1 - x\ln x} + 1 - x^2 \;=\; \frac{x - x^2 + x^3\ln x}{1 - x\ln x} \;=\; \frac{x\,r(x)}{1 - x\ln x},$$

where $r(x) := 1 - x + x^2\ln x$. The denominator exceeds 1, and $r > 0$ on $(0,1)$: indeed $r(1) = 0$, $r(0^+) = 1$, and $r'(x) = x(1 + 2\ln x) - 1 < 0$ on $(0,1)$ because the auxiliary function $u_1(x) := x(1 + 2\ln x)$ satisfies $u_1 \leq 1$ on $(0,1]$, with equality only at $x = 1$ (Lemma A.1 in the Appendix). Hence r decreases strictly from 1 to 0 and is positive. This gives $h > 0$ on $(0,1)$, completing the proof. □

Remark 5.3. Proposition 5.2 says that $\mathcal{T}$ is a (weak) contraction of the whole real line into the positive half-line. In particular the graph of $\mathcal{T}$ is flatter than the diagonal everywhere except at the origin, where it is tangent to it; this is visible in Figure 1 and is the mechanism behind the global dynamics of Section 8. Note the pleasing division of labour in the proof: the elementary inequalities $\mathrm{e}^u \geq 1 + u$ (for $u \geq 0$) and $\mathrm{e}^u \leq 1/(1-u)$ (for $u < 1$) each take care of one side of $x = 1$.

6 Local theory: the Maclaurin expansion and the radius of convergence

The sandwich theorem controls the global position of $\mathcal{T}$; we now zoom in on the origin, where the three normalizations $\mathcal{T}(0) = \mathcal{T}'(0) = 1$, $\mathcal{T}''(0) = 0$ of Proposition 3.5 suggest a degenerate local structure. The degeneracy is real, and it is the entrance to a chain of consequences: an inflection with third-order

contact, a clean convexity dichotomy, and a Maclaurin series whose radius of convergence is set by a pair of complex critical values of f, reached through a picture (Figure 3) that we find to be one of the prettiest by-products of the theory.

6.1 The series

Theorem 6.1 (Maclaurin expansion; the quadratic term vanishes)**.** *The Tahawus function is real-analytic at the origin, and*

$$\begin{aligned}\mathcal{T}(y) \;&=\; 1 + y - \frac{y^3}{2} + \frac{y^4}{6} + \frac{y^5}{2} - \frac{49\,y^6}{120} - \frac{5\,y^7}{9} \\ &+ \frac{4313\,y^8}{5040} + \frac{49\,y^9}{90} - \frac{6341\,y^{10}}{3780} - \frac{11549\,y^{11}}{50400} + \frac{62041117\,y^{12}}{19958400} + O(y^{13}).\end{aligned} \tag{16}$$

In particular the coefficient of y^2 vanishes identically: $\mathcal{T}$ has third-order contact with its tangent line $1 + y$ at the origin. All coefficients are rational numbers.

Proof. Write $u := x - 1$ and recall from (15) that

$$y \;=\; f(1+u) \;=\; u\,\varphi(u), \qquad \varphi(u) \;:=\; \frac{f(1+u)}{u} \;=\; 1 + \frac{u^2}{2} - \frac{u^3}{6} + \frac{u^4}{4} - \frac{7u^5}{40} + \cdots, \tag{17}$$

a power series with $\varphi(0) = f'(1) = 1 \neq 0$. The Lagrange–Bürmann inversion formula applied to $y = u\,\varphi(u)$ gives

$$u(y) \;=\; \sum_{n\geq 1} \frac{y^n}{n}\,[u^{n-1}]\,\varphi(u)^{-n}, \qquad \text{so that} \qquad \mathcal{T}(y) \;=\; 1 + \sum_{n\geq 1} d_n y^n, \quad d_n := \frac{1}{n}[u^{n-1}]\,\varphi(u)^{-n}, \tag{18}$$

with positive radius of convergence, since $\varphi(0) \neq 0$. Each d_n is a rational number, because the coefficients of φ are rational: they are obtained from the exact expansion $E(u) := (1+u)\ln(1+u) = \sum_{k\geq 1}(-1)^{k+1}u^k/(k(k-1))$ by exponentiating ($x^x = \exp E$), dividing by $1+u$, and reverting; all steps are rational recurrences, reproduced in Appendix B, which is how the twelve coefficients displayed in (16) were computed exactly. The vanishing of d_2 was already proved in Proposition 3.5 via $\mathcal{T}''(0) = -f''(1)/f'(1)^3 = 0$; it can also be read off directly from (17), since φ itself has no linear term, so φ^{-2} has no u^1-term. As a numerical check, the partial sum of (16) through y^{12} was evaluated against the directly computed value of $\mathcal{T}$: the differences are 3.2×10^{-11} at $y = 0.15$, 3.3×10^{-7} at $y = 0.3$ and 7.0×10^{-5} at $y = 0.45$, consistent with a twelve-term truncation of a series of radius $R \approx 0.65$ (Subsection 6.3 below). □

Remark 6.2 (why the quadratic term vanishes). The cancellation has a transparent cause in the two-term decomposition (7), $f(x) = x^{x-1} - 1/x$. At $x = 1 + u$,

$$x^{x-1} \;=\; (1+u)^u \;=\; \exp\!\big(u\ln(1+u)\big) \;=\; 1 + 0\cdot u + u^2 - \frac{u^3}{2} + \cdots, \qquad \frac{1}{x} \;=\; 1 - u + u^2 - \cdots,$$

and the two terms have *identical quadratic parts* $+u^2$, which cancel in the difference, while their linear parts differ by exactly u. So f is "linear plus cubic" at $x = 1$, and reversion preserves the pattern: $\mathcal{T}$ is "linear plus cubic" at $y = 0$. Neither e^y nor $1 + y + cy^2$ with $c \neq 0$ shares this property; in the two-parameter family of functions with $\mathcal{T}(0) = \mathcal{T}'(0) = 1$ the Tahawus function sits on the measure-zero slice $c = 0$.

6.2 Convexity dichotomy

The sign of $\mathcal{T}''$ is the mirror of the sign of f'', and f'' has an entirely explicit form which behaves differently on the two sides of $x = 1$.

Theorem 6.3 (convexity dichotomy)**.** *The Tahawus function is strictly convex on* $(-\infty, 0)$ *and strictly concave on* $(0, \infty)$*; equivalently,* f *is strictly concave on* $(0, 1)$ *and strictly convex on* $(1, \infty)$*. The point* $(0, 1)$ *is the unique inflection point of* $\mathcal{T}$*, and there the tangent line* $1 + y$ *has contact of order three.*

Proof. Differentiating (13),

$$f''(x) \;=\; x^{x-1}\Big(\ln x + 1 - \frac{1}{x}\Big)^2 \;+\; \frac{x^x(x+1) - 2}{x^3}. \tag{19}$$

For $x > 1$ we have $x^x > 1$ and $x + 1 > 2$, so the second term is positive and the first is nonnegative: $f'' > 0$.

For $0 < x < 1$, substitute $x = \mathrm{e}^{-s}$, $s > 0$, and write $E := \mathrm{e}^{-s} = x$ and $t := (s-1)E$. Using $\ln x + 1 - \frac{1}{x} = -(1+t)/E$ and $x^{x+2} = \mathrm{e}^{-sE}E^2$, a direct substitution into (19) gives the identity

$$f''(x)\,x^3 \;=\; \Psi(s) - 2, \qquad \Psi(s) \;:=\; \mathrm{e}^{-sE}\Big[(1+t)^2 + 1 + E\Big], \qquad t = (s-1)\mathrm{e}^{-s}, \tag{20}$$

which we verified numerically to fifty digits at several points. Now $(1+t)^2 + 1 + E = 2 + w$ with $w := 2t + t^2 + E$. Since $h(s) := (s-1)\mathrm{e}^{-s} > -1$ for $s > 0$ (Lemma A.2(i) in the appendix), we have $w \geq (1+t)^2 - 1 + E > -1$, so $2 + w > 1 > 0$ and $v := w/2 > -\frac{1}{2}$; the elementary bound $\ln(2+w) = \ln 2 + \ln(1+v) \leq \ln 2 + v$ therefore applies, and

$$\ln\Psi \;=\; -sE + \ln(2+w) \;\leq\; \ln 2 - sE + t + \frac{t^2}{2} + \frac{E}{2} \;=\; \ln 2 + \frac{E}{2}\Big((s-1)^2E - 1\Big) \;<\; \ln 2,$$

the last inequality being Lemma A.2(ii): $(s-1)^2\mathrm{e}^{-s} < 1$ for all $s > 0$. Hence $\Psi < 2$ and $f''(x) < 0$ on $(0, 1)$. Since $\mathcal{T}''(y) = -f''\big(\mathcal{T}(y)\big)/f'\big(\mathcal{T}(y)\big)^3$ and $\mathcal{T}(y) < 1 \iff y < 0$, the claims follow; the inflection statement is Proposition 3.5 plus the sign change just established. □

6.3 The radius of convergence and the slit

The series (16) cannot converge everywhere: $\mathcal{T}$ has finite limits of slope, but the complex continuation of f has critical points, and critical values of f are branch points of $\mathcal{T}$. By (14), the critical points of f are the zeros of

$$G(x) \;:=\; x^{x+1}\Big(\ln x + 1 - \frac{1}{x}\Big) + 1 \qquad \big(= x^2 f'(x)\big), \tag{21}$$

and G has none on the positive real axis (Theorem 3.3). Solving $G = 0$ in the complex plane (principal branch of the power) by path continuation combined with Newton iteration, starting from the direction suggested by the coefficients themselves, locates a conjugate pair of critical points

$$\begin{aligned} x_c &= 0.72822912959789 \;\pm\; 1.14145098012 14\,i, \\ y_c &= f(x_c) \;=\; -0.10489522487779 \;\pm\; 0.64412488108293\,i, \end{aligned}$$

(the signs being paired), at distance

$$R \;:=\; |y_c| \;=\; 0.6526100448\ldots \tag{22}$$

from the origin. A broad sweep of initial conditions for the Newton iteration found no closer critical values; the next nearest lies near the logarithmic singularity of f at $x = 0$, at modulus ≈ 104.9, and plays no role. Two independent cross-checks corroborate (22). First, fitting the dominant conjugate-pair pattern $d_n \sim A\lambda^n + \overline{A}\,\overline{\lambda}^n$ (that is, $d_{n+2} = Sd_{n+1} - Pd_n$) to the sixty exact rational coefficients by least squares yields a singularity estimate of modulus 0.6698, within three percent of (22); naive ratio estimates $|d_n/d_{n+1}|$ are useless here because the dominant singularity is complex, so the ratios oscillate between 0.07 and 7 instead of converging. Second, the partial sums of (16) behave differently on the two sides of R (Table 1): at $0.9R$ the truncation error decreases steadily with the number of terms, while at $1.06R$ it stagnates and oscillates, never falling below forty-five times the best inside value.

Table 1: Partial sums $S_n(y)$ of (16) inside and outside the radius: $|S_n(y) - \mathcal{T}(y)|$. Inside R the error decays geometrically; outside it does not decay at all.

n	$\lvert S_n(0.9R) - \mathcal{T}(0.9R)\rvert$	$\lvert S_n(1.06R) - \mathcal{T}(1.06R)\rvert$
8	1.3×10^{-3}	7.6×10^{-3}
12	2.3×10^{-3}	1.9×10^{-2}
16	1.3×10^{-3}	2.0×10^{-2}
20	5.3×10^{-4}	1.5×10^{-2}
24	1.1×10^{-4}	4.7×10^{-3}

We summarize the state of knowledge as follows: the Maclaurin series of $\mathcal{T}$ has radius of convergence $R = 0.6526100\ldots$, the nearest singularities being the conjugate critical values of f above, at which the inverse function develops square-root branch points. The determination of R is numerical (two independent methods agree), and promoting it to a proof is Open Problem (3).

Figure 3 explains, at the same time, what the complex branch cut of $\mathcal{T}$ looks like. Let γ^+ be the image of the negative real axis under the principal-branch continuation of f,

$$\gamma^+ \;:=\; \big\{\, f(-r) \;:\; r > 0 \,\big\}, \qquad f(-r) \;=\; \frac{(-r)^{-r} - 1}{-r} \;=\; \frac{1 - r^{-r}\mathrm{e}^{-i\pi r}}{r},$$

together with its conjugate $\overline{\gamma^+}$. As $r \to 0^+$ the curve escapes to $-\infty$ at height π, and as $r \to \infty$ it spirals into the origin: the phase $-i\pi r$ rotates faster and faster while the amplitude r^{-r} collapses, so the curve winds through decreasing circles, crossing the real axis exactly at the integer parameters $r = n$, where

$$f(-n) \;=\; \frac{1 - (-1)^n n^{-n}}{n} \;=\; b_n, \tag{23}$$

a *rational* number (Theorem 7.4 below proves these are the only rational points of the slit). The points b_n alternate around $1/n$: $b_n > 1/n$ for n odd, $b_n < 1/n$ for n even, and $b_n \to 0$. The critical values $y_c, \overline{y_c}$ responsible for the radius (22) sit just inside the first turn of the spiral, on either side of the real axis, at the squares marked in the figure; the dotted circle is $|y| = R$.

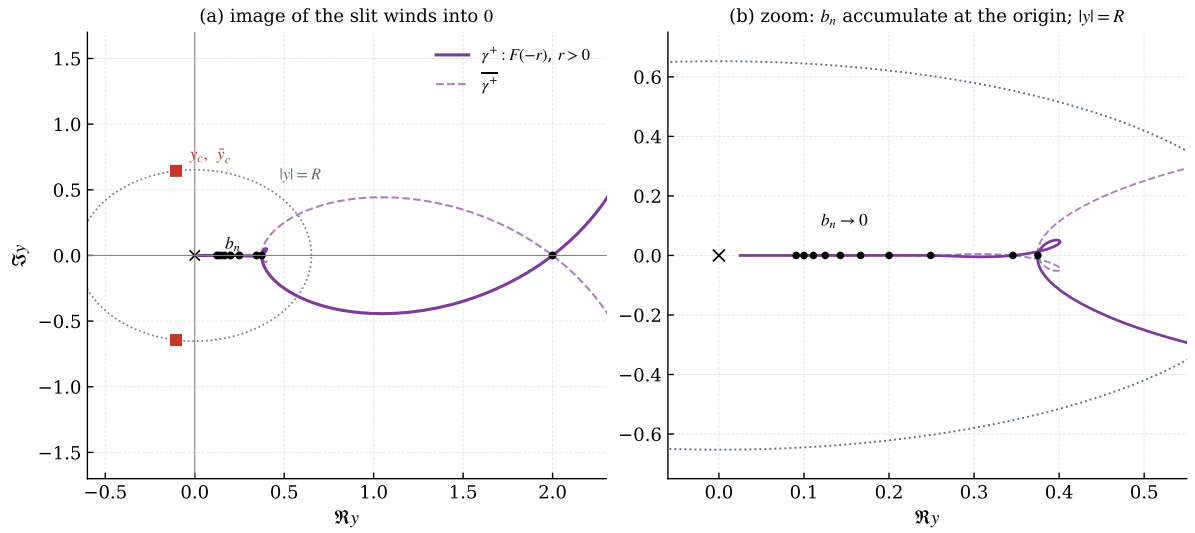


Figure 3: (a) The slit $\gamma^+ = f((-\infty, 0))$ (solid) and its conjugate (dashed): the image of the negative real axis under f spirals from $-\infty + i\pi$ into the origin, crossing the real axis exactly at the rational points $b_n = (1 - (-1)^n n^{-n})/n$ (dots). The critical values y_c, $\overline{y_c}$ (squares) lie at distance $R = 0.6526\ldots$ from the origin (dotted circle) and bound the disk of convergence of the Maclaurin series (16). (b) Zoom near the origin: the crossings b_n accumulate at 0 alternately from both sides of $1/n$.

7 Special values, the golden comparison, and the OEIS constellation

Theorem 4.2 described the arithmetic of $\mathcal{T}(b)$ for rational b in bulk; this section exhibits the concrete specimens. Three families of exact values exist—integers, unit fractions, and negative integers—and around them a constellation of five constants has accumulated in the On-Line Encyclopedia of Integer Sequences, tied together by elementary but pleasing identities.

7.1 Exact values

Proposition 7.1 (exact values)**.** *For every $n \in \mathbb{N}$,*

$$\mathcal{T}\left(\frac{n^n - 1}{n}\right) = n, \qquad \mathcal{T}\left(n^{1-\frac{1}{n}} - n\right) = \frac{1}{n}. \tag{24}$$

The first family starts $0, \frac{3}{2}, \frac{26}{3}, \frac{255}{4}, \frac{3124}{5}, \ldots$ (at $x = 1, 2, 3, 4, 5, \ldots$); the second starts $\sqrt{2} - 2$, $3^{2/3} - 3$, $4^{3/4} - 4 = 2\sqrt{2} - 4, \ldots$ (at $x = \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \ldots$). Together with interpolation, these are the only rational arguments b for which $\mathcal{T}(b)$ is rational or algebraic (Theorem 4.2).

Proof. Both identities are the statement $f(x) = b$ at $x = n$ and $x = 1/n$: indeed $f(n) = (n^n - 1)/n$, and

$$f\left(\frac{1}{n}\right) = \frac{(1/n)^{1/n} - 1}{1/n} = n \cdot \left(n^{-1/n} - 1\right) = n^{1-1/n} - n,$$

where $n^{1-1/n} = (n^{n-1})^{1/n}$ is the surd displayed. The exactness was verified symbolically for $n \le 7$ and numerically to eighty digits. □

The surd family gives $\mathcal{T}$ a supply of algebraic *inputs* with algebraic *outputs*—$\mathcal{T}$ is not a transcendence machine everywhere, only on the rational arguments off the two lattices (24). For instance $\mathcal{T}(\sqrt{2} - 2) = \frac{1}{2}$ exactly, while $\mathcal{T}(-0.6)$, a millimeter to the left on the b-axis, is transcendental.

Table 2: The Tahawus root versus the metallic root of the same super-quadratic/quadratic b. The difference vanishes exactly at $b = 0$ and $b = 3/2$ and changes sign there.

b	$\mathcal{T}(b)$	φ_b	$\mathcal{T}(b) - \varphi_b$
−1	0.3036591	0.6180340	−0.3143749
−0.5	0.5565737	0.7807764	−0.2242027
0.5	1.4559593	1.2807764	+0.1751828
1	1.7767750	1.6180340	+0.1587411
1.4	1.9609108	1.9206556	+0.0402552
1.49	1.9961956	1.9920064	+0.0041892
1.5	2.0000000	2.0000000	0
1.51	2.0037821	2.0080064	−0.0042243
2	2.1657594	2.4142136	−0.2484542
3	2.4019926	3.3027756	−0.9007830
10	3.0770622	10.0990195	−7.0219573

7.2 The golden comparison

The quadratic pencil $x^2 = bx + 1$ has positive root $\varphi_b = (b + \sqrt{b^2+4})/2$, the family of metallic means containing the golden ratio $\varphi = \varphi_1$. The super-quadratic pencil has positive root $\mathcal{T}(b)$. The comparison between the two turns out to be a *single sign computation*.

Theorem 7.2 (golden comparison)**.** *For all real b,*

$$\mathcal{T}(b) > \varphi_b \iff 0 < b < \frac{3}{2}, \qquad \mathcal{T}(b) = \varphi_b \iff b \in \Big\{0, \frac{3}{2}\Big\},$$
$$\mathcal{T}(b) < \varphi_b \iff b < 0 \text{ or } b > \frac{3}{2}.$$

The two exact coincidences are $\mathcal{T}(0) = \varphi_0 = 1$ and $\mathcal{T}(\frac{3}{2}) = \varphi_{3/2} = 2$.

Proof. Since f is strictly increasing, $\mathcal{T}(b) > \varphi_b$ is equivalent to $f(\varphi_b) < b$. But φ_b solves $x^2 = bx + 1$, i.e. $b = \varphi_b - 1/\varphi_b$; therefore

$$f(\varphi_b) - b \;=\; \Big(\varphi_b^{\varphi_b - 1} - \frac{1}{\varphi_b}\Big) - \Big(\varphi_b - \frac{1}{\varphi_b}\Big) \;=\; \varphi_b^{\varphi_b - 1} - \varphi_b \;=\; \varphi_b\big(\varphi_b^{\varphi_b - 2} - 1\big),$$

whose sign is the sign of $(\varphi_b - 2)\ln\varphi_b$. As $b \mapsto \varphi_b$ is an increasing bijection $\mathbb{R} \to (0, \infty)$ with $\varphi_0 = 1$ and $\varphi_{3/2} = 2$, the sign is positive exactly for $1 < \varphi_b < 2$, i.e. $0 < b < \frac{3}{2}$, zero exactly at $\varphi_b \in \{1, 2\}$, and negative on the two outside intervals. □

Remark 7.3. The coincidence set $\{0, \frac{3}{2}\}$ reflects the accident $n^n = n^2$, which holds *only* for $n \in \{1, 2\}$: the super-square and the square agree at 1 and 2 and nowhere else. It is pleasant that the two most famous members of the two pencils—the golden ratio and the Tahawus constant—obey $\varphi < \alpha = \mathcal{T}(1) = 1.7767\ldots$, and that the "excess" $\mathcal{T}(b) - \varphi_b$, which vanishes at both ends of $(0, \frac{3}{2})$, peaks at about $0.19000\ldots$ near $b \approx 0.7$ (between the rows 0.5 and 1 of Table 2).

7.3 Negative integer roots, and the census of real roots

For negative x the power x^x is multivalued; we use the principal branch $x^x = \exp(x \operatorname{Log} x)$, under which x^x is real precisely for negative integers (this is the branch plotted in Figure 3). The defining equation of the super-quadratic then admits a second, exactly solvable family.

Theorem 7.4 (negative integer roots)**.** (i) *For every $n \in \mathbb{N}$, the negative integer $x = -n$ solves*

$$x^x \;=\; b_n\, x + 1 \qquad \textit{with} \qquad b_n \;:=\; \frac{1 - (-1)^n\, n^{-n}}{n} \;\in\; \mathbb{Q},$$

e.g. $(-1)^{-1} = -1 = 2\cdot(-1)+1$ *(so* $b_1 = 2$*),* $(-2)^{-2} = \frac{1}{4} = \frac{3}{8}\cdot(-2)+1$*, and the sequence* $b_n = 2, \frac{3}{8}, \frac{28}{81}, \frac{255}{1024}, \ldots$ *alternates around* $1/n$ *and tends to* 0*.*

(ii) *On the principal branch,* $f(-r)$ *is real* if and only if $r \in \mathbb{N}$*; hence the points* (23) *are exactly the real points of the slit* $\gamma^+ \cup \overline{\gamma^+}$ *of Figure 3.*

(iii) α *is the unique* real *root of* $x^x = x+1$ *in either convention for negative powers (principal branch, or the real-branch convention* $(-p/q)^{-p/q} := \big((-p/q)^{-p}\big)^{1/q}$ *with* q *odd).*

Proof. (i) $(-n)^{-n} = (-1)^n n^{-n}$, and $(-1)^n n^{-n} = b_n\cdot(-n)+1$ rearranges to $b_n = (1-(-1)^n n^{-n})/n$; these are rational, and the identities were verified symbolically for $n \le 7$.

(ii) On the principal branch $(-r)^{-r} = \exp\big(-r\ln r - i\pi r\big) = r^{-r}\mathrm{e}^{-i\pi r}$, so

$$f(-r) \;=\; \frac{(-r)^{-r}-1}{-r} \;=\; \frac{1-r^{-r}\mathrm{e}^{-i\pi r}}{r},$$

which is real iff $\mathrm{e}^{-i\pi r}$ is real, i.e. iff $\sin(\pi r) = 0$, i.e. iff $r \in \mathbb{Z}$; positivity of r gives $r \in \mathbb{N}$. Then (23) applies.

(iii) On the principal branch, $x^x \in \mathbb{R}$ for $x < 0$ only at the integers (part (ii)), and $x = -n$ solves $x^x = x+1$ only if $b_n = 1$, which never happens: $b_1 = 2$, and for $n \ge 2$, $b_n \le (1+2^{-2})/n \le \frac{5}{4}\cdot\frac{1}{2} < 1$. Under the real-branch convention, a negative candidate is $x = -p/q$ in lowest terms with q odd. If $q \ge 2$, then $x^x = (-1)^p(q/p)^{p/q}$ is *irrational*: were $(q/p)^{p/q} = m/n$ rational, raising to the q-th power would give $q^p n^q = m^q p^p$, and the valuation argument of Lemma 4.1 applied with p and q interchanged (any prime divisor ℓ of q divides neither p nor m^q's factor, forcing $q \mid v_\ell(q) \le \log_2 q < q$, a contradiction) rules this out. An irrational left-hand side cannot equal the rational $x+1$, so no $q \ge 2$ works; $q = 1$ returns to the integers, already excluded. Hence the only real root of $x^x = x+1$ is the positive one, α. □

7.4 The constellation of constants

Around α and $\beta = \mathcal{T}(-1)$ there orbits a small family of constants, all present in the OEIS, all transcendental, and all connected by elementary transformations. Recall $\beta^\beta + \beta = 1$ and put

$$c \;:=\; \frac{1}{\beta} \;=\; 3.2931662874118610\ldots, \qquad v \;:=\; c-1 \;=\; \frac{1}{\beta}-1 \;=\; 2.2931662874118610\ldots$$

Theorem 7.5 (the constellation)**.** (i) *The following are equivalent for* $v > 0$ *(and all hold for* $v = 1/\beta - 1$*):*

$$v^{\,1+1/v} = v+1 \iff v = \Big(1+\frac{1}{v}\Big)^v \iff v^{\,1/v} = (v+1)^{\,1/(v+1)} \iff v^{v+1} = (v+1)^v.$$

(ii) $c = v+1$ *satisfies* $c^{\,c-1} = (c-1)^c$*, equivalently* $\frac{\ln c}{c} = \frac{\ln(c-1)}{c-1}$*; this equation has a unique real solution, lying in* $(\mathrm{e}, \mathrm{e}+1)$*.*

(iii) *The equation* $v^{\,1/v} = (v+1)^{\,1/(v+1)}$ *has the unique solution* $v \in (1,\mathrm{e})$ *above:* v *and* $v+1$ *are the unique pair of* consecutive *positive reals with equal* $x^{1/x}$*, the analogue for* $x^{1/x}$ *of the classical pair* $(2,4)$*.*

(iv) *For each integer* $k \ge 1$ *the equation* $x^{\,1+1/x} = x+k$ *has a unique positive solution* $\upsilon_k > 1$*;* $\upsilon_1 = v$ *and*

$$\upsilon_2 \;=\; 5.5079856527731782575890\ldots \qquad (A144184).$$

(v) *All five constants* α *(A124930),* β *(A226568),* $c = 1/\beta$ *(A169862),* $v = c-1$ *(A085846) and* υ_2 *(A144184) are transcendental.*

Proof. (i) $v^{\,1+1/v} = v+1$ says $v \cdot v^{1/v} = v+1$, i.e. $v^{1/v} = (v+1)/v = 1+1/v$; raising both sides to the power v gives $v = (1+1/v)^v$. Next, $v^{1/v} = (v+1)^{1/(v+1)}$ raised to the power $v(v+1)$ gives $v^{v+1} = (v+1)^v$, and dividing the exponents by $v(v+1)$ returns the equality of the $1/x$-powers; and $v^{v+1} = (v+1)^v$ divided by v^v is $v = ((v+1)/v)^v = (1+1/v)^v$. For the bridge from β: $\beta^\beta = 1-\beta$ with $\beta = 1/c$ reads $(1/c)^{1/c} = (c-1)/c$; raising to the power c gives $1/c = (c-1)^c/c^c$, i.e. $c^{c-1} = (c-1)^c$, which is (ii) for c, and it translates to $v^{1/v} = (v+1)^{1/(v+1)}$ by taking $c = v+1$ and dividing exponents by $c(c-1)$.

(ii) The function $\ln x/x$ increases on $(0,\mathrm{e})$ and decreases on (e,∞); thus $\ln c/c = \ln(c-1)/(c-1)$ forces $c-1 < \mathrm{e} < c$, i.e. $c \in (\mathrm{e}, \mathrm{e}+1)$, and on this interval the difference $\ln x/x - \ln(x-1)/(x-1)$ is strictly decreasing (its first term decreases, its second increases), vanishing at most once; it vanishes at $c = 1/\beta$ by the computation above.

(iii) is (ii) restated at $x = v+1$: the pairs $x, x+1$ with equal $x^{1/x}$ correspond exactly to $c = x+1$ solving (ii), and $x = v \in (1,\mathrm{e})$.

(iv) Put $\psi(x) := x^{1+1/x} - x = x\big(x^{1/x} - 1\big)$. Then $\psi(1) = 0$, and ψ is strictly increasing on $[1,\infty)$: indeed, with $L := \ln x \geq 0$,

$$x^{1/x-1}\big(x+1-L\big) \;=\; \frac{x^{1/x}}{x}\big(x+1-L\big) \;\geq\; \Big(1+\frac{L}{x}\Big)\frac{x+1-L}{x} \;=\; \frac{(x+L)(x+1-L)}{x^2} \;>\; 1,$$

the last step because $(x+L)(x+1-L) = x^2+x+L-L^2 > x^2$, owing to $x > L^2 - L$, which follows from $L^2 < x$ (Lemma A.3); and the chain says exactly that $\psi'(x) = x^{1+1/x}\big(x+1-\ln x\big)/x^2 - 1 > 0$. Since $\psi(1) = 0$ and $\psi(x) \to \infty$ (as $x^{1/x} \to 1$ and $x \to \infty$), each value $k > 0$ is taken exactly once. The value of υ_2 was located numerically and its equation verified.

(v) α and β are transcendental by Corollary 4.3; c and v are nonconstant rational functions of β, hence transcendental. For $\upsilon_2 =: w$: if w were algebraic irrational, then $w \notin \{0,1\}$ and $w^{1/w}$ would be transcendental by Gelfond–Schneider, making $w^{\,1+1/w} = w \cdot w^{1/w}$ transcendental—contradicting $w^{\,1+1/w} = w+2$ algebraic. If $w = p/q$ were rational (lowest terms), raising $(p/q)^{1+q/p} = (p+2q)/q$ to the p-th power gives

$$p^{\,p+q} \;=\; (p+2q)^p\, q^{\,q}.$$

Any odd prime $\ell \mid p$ divides neither q nor $p+2q$ (since $\gcd(p, p+2q)$ divides 2), so comparing v_ℓ on both sides gives $(p+q)\, v_\ell(p) = 0$, impossible; hence $p = 2^a$. If $a \geq 2$, then $v_2(p+2q) = 1$ (as $2^{a-1}+q$ is odd) and v_2 gives $a(p+q) = p$, impossible for $a \geq 2$. If $a = 1$, then $p = 2$ and $v_2(2+2q) = 1+v_2(1+q)$, so $(2+q) = 2(1+v_2(1+q))$, forcing the odd number q to equal the even number $2\,v_2(1+q)$—impossible. So w is neither rational nor algebraic irrational. □

Remark 7.6 (credits and digit agreement). The five OEIS entries are A124930 (α, root of $x^x = x+1$), A226568 (β, "the number defined by $x^x + x = 1$"), A169862 (c, root of $x^{x-1} = (x-1)^x$), A085846 (v, root of $x = (1+1/x)^x$) and A144184 (υ_2, root of $x^{1+1/x} = x+2$) [3–7]. Our independently computed decimal expansions of α and v agree digit for digit with the seventy-digit strings recorded in A124930 and A085846, and β agrees with the blog post's value [1]; $c = v+1$ and υ_2 were verified through their defining equations. Lord [12] proved the transcendence of v by the equivalent equation $x = (1+1/x)^x$; Theorem 7.5(v) extends the pattern to the whole family υ_k. The quantity $\alpha^{\alpha-1} = 1+1/\alpha$ implicit in the figure, and the identity $\beta^{\beta-1} = (1-\beta)/\beta = v$, are the two edges along which the constellation connects back to $\mathcal{T}$.

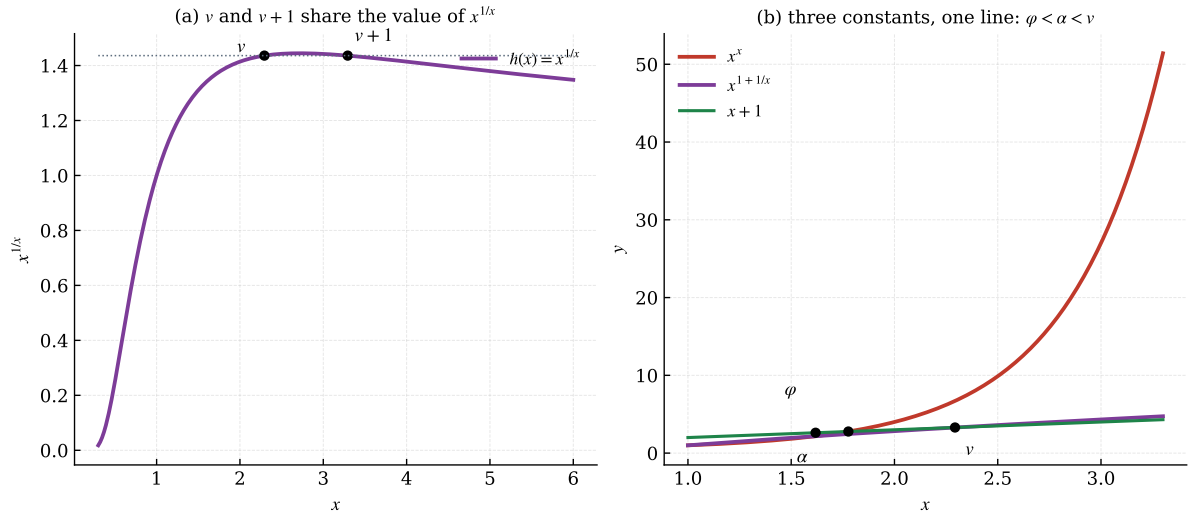


Figure 4: (a) The map $h(x) = x^{1/x}$ is unimodal with maximum at e; the consecutive pair $v, v+1$ (marked, with the dotted level line) is the unique pair of consecutive reals on which h takes equal values—for h, what the pair $(2, 4)$ is classically. (b) The line $y = x + 1$ (green), the curve x^x (red) and the curve $x^{1+1/x}$ (purple): the two curves exchange order at $x = \varphi$ (where the exponents x and $1 + 1/x$ coincide), the line is met by x^x at α and by $x^{1+1/x}$ at v; consequently $\varphi < \alpha < v$, the three parameters marked at their heights on the line.

8 Dynamics: the attracting fixed point

Because $0 < \mathcal{T}'(y) \leq 1$ everywhere with equality only at the origin (Proposition 5.2), the Tahawus function is a weak contraction of $\mathbb{R}$ into $(0, \infty)$, and its iteration theory is as clean as one could hope: there is a single fixed point, it attracts everything, and every orbit approaches it monotonically.

Theorem 8.1 (fixed point)**.** (i) *$\mathcal{T}$ has a unique fixed point*

$$\delta \;=\; 2.22883310327811897061876155 46\ldots, \qquad \delta^{\,\delta} \;=\; \delta^2 + 1,$$

which is the unique positive root of $x^x = x^2 + 1$ and lies in $(2, 3)$.

(ii) *δ is transcendental.*

(iii) *Every orbit $y_{n+1} = \mathcal{T}(y_n)$ converges to δ: strictly increasing if $y_0 < \delta$, strictly decreasing if $y_0 > \delta$ (after one step from any $y_0 \leq 0$).*

(iv) *The fixed point is attracting, with $\mathcal{T}'(\delta) = 0.2615\ldots < 1$.*

Proof. (i) For $y > 0$, $\mathcal{T}(y) = y \iff y^y - 1 = y^2 \iff h(y) := y^y - y^2 - 1 = 0$; and no $y \leq 0$ is fixed since $\mathcal{T}(y) > 0 \geq y$ with $\mathcal{T}(0) = 1$. Now $h < 0$ on $(0, 2]$: for $0 < x < 1$, $x^x < 1 < 1 + x^2$; for $1 \leq x \leq 2$, $x^{x-2} \leq 1$ gives $x^x \leq x^2 < x^2 + 1$. On $(2, \infty)$ write $h(x) = x^2\big(x^{x-2} - 1 - x^{-2}\big)$, whose sign is the sign of

$$K(x) := (x - 2)\ln x - \ln\big(1 + x^{-2}\big), \qquad K'(x) = \ln x + \frac{x-2}{x} + \frac{2}{x(x^2+1)} \;>\; 0 :$$

each term is positive for $x > 2$. Since $K(2) = -\ln\frac{5}{4} < 0 < \ln 3 - \ln\frac{10}{9} = K(3)$, h has exactly one zero $\delta \in (2, 3)$.

(iii) $\mathcal{T}(y) > y \iff h(y) < 0$ (apply the increasing f to both sides: $\mathcal{T}(y) > y \iff y = f(\mathcal{T}(y)) > f(y)$, and $f(y) - y = h(y)/y$). So $\mathcal{T}(y) > y$ on $(0, \delta)$ and $\mathcal{T}(y) < y$ on (δ, ∞). As $\mathcal{T}$ is increasing with $\mathcal{T}(\delta) = \delta$, it maps $(0, \delta)$ into $(0, \delta)$ and (δ, ∞) into (δ, ∞); monotone bounded sequences converge, and continuity forces the limit to be a fixed point. For $y_0 \leq 0$, $y_1 = \mathcal{T}(y_0) \in (0, 1] \subset (0, \delta)$.

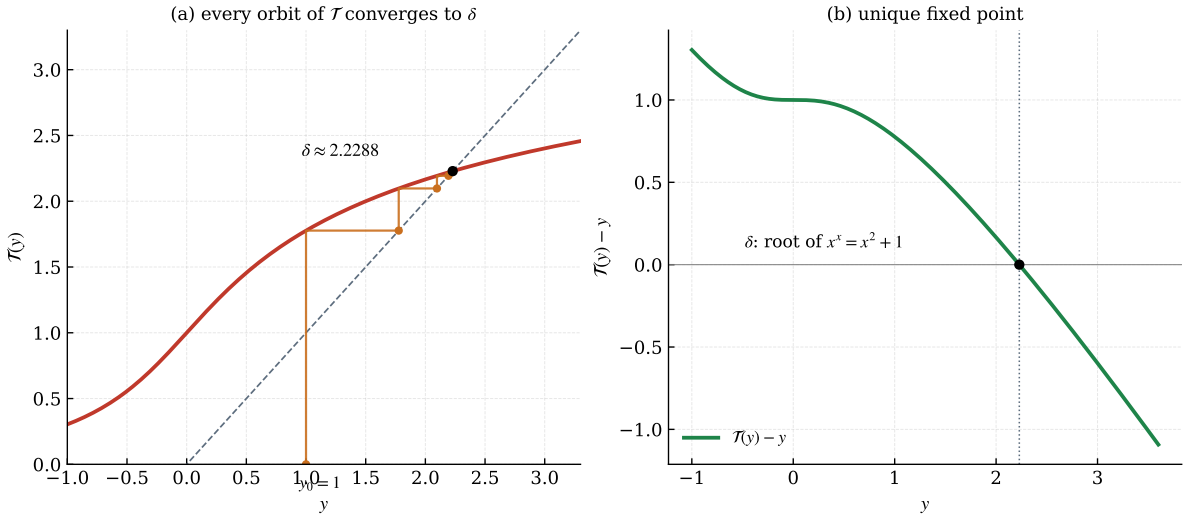


Figure 5: (a) Staircase diagram of the orbit of $\mathcal{T}$ started at $y_0 = 1$ (orange): the orbit climbs monotonically to the attracting fixed point $\delta \approx 2.2288$ (black dot), where the graph of $\mathcal{T}$ (red) crosses the diagonal (dashed). (b) The displacement $\mathcal{T}(y) - y$: positive exactly on $(0, \delta)$, vanishing at the unique fixed point, root of $x^x = x^2 + 1$.

(ii) If δ were an algebraic irrational, Gelfond–Schneider would make δ^δ transcendental, contradicting $\delta^\delta = \delta^2 + 1$ algebraic. If $\delta = p/q$ (lowest terms, $p > q$ since $\delta > 2$), then raising $(p/q)^{p/q} = (p^2+q^2)/q^2$ to the q-th power gives $p^p\, q^{2q-p} = (p^2+q^2)^q$. But $\gcd\big(p,\, p^2+q^2\big) = \gcd(p, q^2) = 1$, so any prime $\ell \mid p$ has $v_\ell(\text{LHS}) = p\, v_\ell(p) > 0$ (as $\ell \nmid q$) while $v_\ell(\text{RHS}) = 0$: impossible. Hence $p = 1$, contradicting $p > q \geq 1$. So δ is transcendental.

(iv) $\mathcal{T}'(\delta) = \delta^2 / f'(\delta)$ with $f'(\delta) = \delta^{\delta-1}\big(\ln\delta + 1 - \frac{1}{\delta}\big) + \delta^{-2} = 9.382\ldots$, giving $\mathcal{T}'(\delta) = 0.2615435596\ldots$; it is < 1 by Proposition 5.2. □

The orbit of 1 is a good advertisement for the speed of the attraction, and for how quickly the iterates forget that they started at the Tahawus constant:

$$1,\ 1.7767750401,\ 2.0971558587,\ 2.1933551737,\ 2.2194806484,$$
$$2.2263819619,\ 2.2281916756,\ 2.2286653182,\ 2.2287892186,\ 2.2288216254,\ \ldots \to \delta,$$

each step multiplying the error by roughly $\mathcal{T}'(\delta) = 0.2615$. An orbit started at $y_0 = -5$ reaches 0.00622, then 1.00622, then follows the same staircase: one step suffices to enter $(0, \delta)$, as promised. Newton's method on $g(x) = x^x - x^2 - 1$, started at 2, exhibits its usual quadratic convergence: 2.36067, 2.25177, 2.22964, 2.22883, 2.228833103278118971. We remark that δ is the $n = 2$ member of the family of *super-golden* constants {roots of $x^x = x^n + 1$}, whose first member ($n = 1$) is α itself; see Open Problem (5).

8.1 The nested composition of the blog post, made convergent

The infinite composition (6) is not a closed form, but it *is* a numerical algorithm: read from the outside in, it is the fixed-point iteration

$$x_{n+1} = \operatorname{ssqrt}(x_n + 1) = \mathrm{e}^{W(\ln(x_n+1))}, \qquad x_0 = \operatorname{ssqrt}(2) = \mathrm{e}^{W(\ln 2)}, \tag{25}$$

where the starting value is the first approximant obtained by ignoring the "+1" (this is exactly the nesting displayed in the original post). The remarkable fact is that this iteration is a *contraction with a closed-form rate*, in sharp contrast with the power-tower iteration of the next section.

Proposition 8.2 (the W-iteration converges geometrically)**.** *Let* $g(x) := \operatorname{ssqrt}(x+1) = \mathrm{e}^{W(\ln(x+1))}$ *on* $[0, \infty)$. *Then* g *is increasing,* $g(x) > x$ *on* $[0, \alpha)$, $g(x) < x$ *on* (α, ∞), *and every orbit of* (25) *converges*

monotonically to α. Moreover g is differentiable at α with

$$g'(\alpha) \;=\; \frac{1}{(1+\alpha)\big(1+\ln\alpha\big)} \;=\; 0.2286830163\ldots \;<\; 1, \tag{26}$$

so the convergence is geometric with ratio (26)*.*

Proof. Monotonicity of g is that of $x \mapsto x{+}1, \ln, W$ and exp. For $x \in [0,1)$ we have $g(x) \geq \operatorname{ssqrt}(1) = 1 > x$. For $x \geq 1$ (where $t \mapsto t^t$ is increasing), $g(x) > x \iff x^x < x+1 \iff x < \alpha$; this uses the (known) fact that $x^x - x - 1$ is strictly increasing on $[1,\infty)$, with root α. Hence g maps $[0,\alpha]$ into $[1,\alpha]$ and (α,∞) into (α,∞) with $g(x) < x$ there, and monotone convergence proceeds exactly as in Theorem 8.1(iii), the only fixed point of g in $[0,\infty)$ being α. For the rate: $g'(x) = g(x)\,W'(\ln(x+1))\,/(x+1)$ with $W'(z) = W(z)/\big(z(1+W(z))\big)$; at $x=\alpha$ we have $g(\alpha)=\alpha$ and $W(\ln(\alpha+1)) = \ln\alpha$, because $(\ln\alpha)\,\mathrm{e}^{\ln\alpha} = \alpha\ln\alpha = \ln(\alpha^\alpha) = \ln(\alpha+1)$. Substituting $z = \ln(\alpha+1) = \alpha\ln\alpha$,

$$g'(\alpha) \;=\; \alpha\cdot\frac{\ln\alpha}{\alpha\ln\alpha\,(1+\ln\alpha)}\cdot\frac{1}{\alpha+1} \;=\; \frac{1}{(1+\alpha)(1+\ln\alpha)}. \qquad \square$$

The first iterates of (25) are

$$1.55961,\; 1.7245718243,\; 1.7646972820,\; 1.7740056666,\; 1.7761413436,\; \ldots,\; 1.7767750401 \;=\; \alpha,$$

reaching α to nine significant digits in eight steps; the observed error ratios 0.23136, 0.22930, 0.22882, 0.22872, 0.22869, 0.228685, … march to the closed form (26) exactly. This is the honest content of the blog post's nested expression: not a closed form, but a rapidly convergent evaluation scheme. The same scheme works verbatim for general b, iterating $x \mapsto \mathrm{e}^{W(\ln(bx+1))}$.

9 The infinite power tower, corrected

The original post closes with a flourish: since

$$\alpha^\alpha \;=\; \alpha+1, \qquad \text{i.e.} \qquad \alpha^\alpha - 1 \;=\; \alpha, \tag{27}$$

the constant α can be substituted into itself forever,

$$\alpha \;=\; \alpha^\alpha - 1 \;=\; \alpha^{\alpha^\alpha - 1} - 1 \;=\; \alpha^{\alpha^{\alpha^\alpha - 1} - 1} - 1 \;=\; \cdots,$$

an identity of infinite nested exponentials with a "-1" attached at every level. As an *algebraic* identity the chain is unimpeachable: the sequence $x_{n+1} = \alpha^{x_n} - 1$ started at $x_0 = \alpha$ is exactly constant. As a statement about a *limit*, however—the natural reading of an infinite expression—it is wrong, and the correction is instructive.

Theorem 9.1 (the decremented tower)**.** *Let $1 < a < \mathrm{e}$, let $g_a(x) := a^x - 1$, and let $x_+(a) > 0$ denote the unique positive solution of $a^x = x+1$ (it exists and is unique). Then:*

(i) *g_a has exactly two nonnegative fixed points, 0 and $x_+(a)$.*

(ii) *The fixed point 0 is attracting, with multiplier $g_a'(0) = \ln a < 1$; the fixed point $x_+(a)$ is* repelling*: its multiplier satisfies*

$$g_a'\big(x_+(a)\big) \;=\; \big(x_+(a)+1\big)\ln a \;=\; \frac{x_+(a)+1}{x_+(a)}\,\ln\big(x_+(a)+1\big) \;>\; 1, \tag{28}$$

by the inequality $\ln(1+u) > u/(1+u)$ for $u > 0$.

(iii) *Every orbit started in $(-1, x_+(a))$ converges* monotonically *to 0; every orbit started above $x_+(a)$ diverges monotonically to $+\infty$; the orbit started exactly at $x_+(a)$ is stationary.*

For the Tahawus constant, $a = \alpha \in (1, \mathrm{e})$ and $x_+(\alpha) = \alpha$ (this is (27)), so the decremented tower built on α converges to 0 from every starting point except α itself, where it is stationary, and above which it diverges.

Proof. Put $h_a(x) := a^x - x - 1 = g_a(x) - x$. Then $h_a(0) = 0$, $h_a'' = (\ln a)^2 a^x > 0$, and $h_a'(x) = \ln(a)\, a^x - 1$ is strictly increasing with $h_a'(0) = \ln a - 1 < 0$ and $h_a' \to \infty$. So h_a decreases from $h_a(0) = 0$ on $[0, x^*]$ (where x^* is the zero of h_a'), hence is negative on $(0, x^*]$, and then increases to $+\infty$, crossing zero exactly once more at $x_+(a) > x^*$. On $(-1, 0)$: h_a is decreasing there (as $(-1, 0) \subset (-\infty, x^*)$) with $h_a(0) = 0$, so $h_a > 0$ on $(-1, 0)$: no negative fixed points; and points $x \le -1$ are mapped into $(-1, 0)$ in one step since $g_a(x) \in (-1, a^{-1} - 1] \subset (-1, 0)$. The sign of $g_a(x) - x = h_a(x)$ is thus $+, -, +$ on the intervals $(-1, 0)$, $(0, x_+)$, (x_+, ∞), and g_a maps each of the first two intervals into itself, giving monotone convergence to the unique fixed point of each interval, namely 0; on the third interval the orbit increases without fixed points above, so it diverges. The multipliers are $g_a'(x) = \ln(a)\, a^x$: at 0 this is $\ln a < 1$; at x_+, using $a^{x_+} = x_+ + 1$, $g_a'(x_+) = (x_+ + 1) \ln a = \frac{x_+ + 1}{x_+} \ln(x_+ + 1) > 1$ by $\ln(1 + u) > \frac{u}{1+u}$. □

For $a = \alpha$ the multipliers come out as

$$g_\alpha'(0) \;=\; \ln \alpha \;=\; 0.5747999458\ldots, \qquad g_\alpha'(\alpha) \;=\; (1+\alpha)\ln\alpha \;=\; 1.5960901426\ldots \;>\; 1,$$

and the orbit started at $x_0 = \frac{1}{2}$ tells the whole story:

$$0.3330,\ 0.2109,\ 0.1289,\ 0.0769,\ 0.0452,\ 0.0263,\ 0.0152,\ 0.0088,\ 0.0051,\ 0.0029,\ \ldots \;\to\; 0,$$

with successive ratios $0.633, 0.611, 0.597, \ldots$ settling toward the predicted $\ln \alpha = 0.5748$; by step 29 the iterate is 4.5×10^{-8}. So the corrected reading of the post's identity is:

> *The decremented tower of base α converges—but to 0, not to α. The value α is reachable only by the exactly stationary (and repelling) orbit, which is an algebraic identity, not a limit process.*

Two footnotes complete the picture. First, the plain (undecremented) tower $\alpha^{\alpha^{\alpha^{\cdots}}}$ does not converge at all: Euler's criterion [10] requires the base to lie in $[\mathrm{e}^{-\mathrm{e}}, \mathrm{e}^{1/\mathrm{e}}]$, and $\alpha = 1.7767\ldots$ exceeds $\mathrm{e}^{1/\mathrm{e}} = 1.4446678\ldots$. The decrements do tame the divergence, but the tamed limit is 0. Second, the contrast between Sections 8.1 and this one is stark and practical: *the same fixed point α* is repelling for the map $x \mapsto \alpha^x - 1$ (multiplier 1.596) and attracting for the map $x \mapsto \mathrm{e}^{W(\ln(x+1))}$ (multiplier 0.229), which are algebraically the same equation. The nest of the blog post should always be evaluated through W, never through the power.

Remark 9.2 (general bases). Theorem 9.1 is stated for the Tahawus constant but proved for every base $1 < a < \mathrm{e}$: the pair of fixed points $\{0, x_+(a)\}$ always exists, 0 always attracts, and $x_+(a)$ always repels, with (28) holding for every a. For $a \ge \mathrm{e}$ even the fixed point 0 loses stability (multiplier $\ln a \ge 1$) and no positive fixed point exists. The Tahawus constant is simply the special base a for which the repelling fixed point happens to *coincide with the base itself*—the self-referential charm that motivated the original post's identity (27).

10 Asymptotics

Both ends of the real line have clean asymptotics, and the right end hides a surprise: at infinity the Tahawus function is a square super-root *shifted by exactly one*. Throughout, $x := \mathcal{T}(y)$, and the starting point is an *exact* identity obtained by taking logarithms of the defining equation $x^x = xy + 1$ and peeling off $\ln x$:

$$x \ln x \;=\; \ln x + \ln y + \ln\Big(1 + \frac{1}{xy}\Big) \qquad \Longleftrightarrow \qquad (x-1)\ln x \;=\; \ln y + \ln\Big(1 + \frac{1}{xy}\Big), \tag{29}$$

the *master identity on the right*. The super square root $\mathrm{ssqrt}(y) = \mathrm{e}^{W(\ln y)}$ of (4) is the solution of $x \ln x = \ln y$; the Tahawus function is the solution of (29), which differs from it by the shift $x \mapsto x - 1$ in

the left factor and by a logarithmically tiny correction on the right. Both differences push x *up* relative to $\mathrm{ssqrt}(y)$.

Theorem 10.1 (asymptotics). (i) (far left) *As* $y \to -\infty$,

$$\ln \mathcal{T}(y) \;=\; y - \frac{y^2}{2}\,\mathrm{e}^{y}\big(1+o(1)\big), \qquad \textit{equivalently} \qquad \mathcal{T}(y) \;=\; \mathrm{e}^{y} - \frac{y^2}{2}\,\mathrm{e}^{2y}\big(1+o(1)\big). \tag{30}$$

(ii) (far right, plus-one law) *As* $y \to +\infty$,

$$\mathcal{T}(y) \;-\; \mathrm{ssqrt}(y) \;\longrightarrow\; 1. \tag{31}$$

(iii) (far right, logarithmic scale) *As* $y \to +\infty$,

$$\mathcal{T}(y) \;=\; \frac{\ln y}{\ln\ln y}\Big(1 + \frac{\ln\ln\ln y}{\ln\ln y} + O\Big(\big(\tfrac{\ln\ln\ln y}{\ln\ln y}\big)^2\Big)\Big). \tag{32}$$

Proof. (i) Expand f at $x = 0^+$: with $u := x\ln x \to 0$,

$$y \;=\; \frac{\mathrm{e}^{u}-1}{x} \;=\; \frac{u + \frac{u^2}{2} + O(u^3)}{x} \;=\; \ln x + \frac{x\ln^2 x}{2} + O\big(x^2\ln^3 x\big).$$

Since $x\ln^2 x \to 0$, we get $y - \ln x \to 0$, i.e. $\ln x = y + o(1)$ and $x = \mathrm{e}^{y}(1+o(1))$. Substituting back,

$$y - \ln x \;=\; \frac{x\ln^2 x}{2}\big(1+o(1)\big) \;=\; \frac{y^2}{2}\,\mathrm{e}^{y}\big(1+o(1)\big),$$

which is (30); the second form follows from $\exp\big(-\frac{y^2}{2}\mathrm{e}^{y}(1+o(1))\big) = 1 - \frac{y^2}{2}\mathrm{e}^{y}(1+o(1))$, legitimate because $\frac{y^2}{2}\mathrm{e}^{y} \to 0$.

(ii) Write $L := \ln y$, let $x_0 := \mathrm{ssqrt}(y) = \mathrm{e}^{W(L)}$ (so $x_0\ln x_0 = L$), and let $\Phi(t) := (t-1)\ln t$, strictly increasing on $[1,\infty)$. By (29), $x = \Phi^{-1}(L+\varepsilon)$ with $\varepsilon := \ln\big(1+\frac{1}{xy}\big) \to 0$, while $\Phi(x_0+1) - \Phi(x_1) = \Phi(x_0+1) - L$ where $x_1 := \Phi^{-1}(L)$. Now

$$\Phi(x_0+1) - L \;=\; x_0\ln(x_0+1) - x_0\ln x_0 \;=\; x_0\ln\Big(1+\frac{1}{x_0}\Big) \;\longrightarrow\; 1,$$

since $x_0 \to \infty$. By the mean value theorem there is η between x_1 and x_0+1 with $(x_0+1-x_1)\,\Phi'(\eta) = \Phi(x_0+1) - L \to 1$; as $x_0 \to \infty$ forces $\eta \to \infty$ and $\Phi'(\eta) = \ln\eta + 1 - \frac{1}{\eta} \to \infty$, we conclude $x_1 - x_0 \to 1$. Similarly $x - x_1 = \Phi^{-1}(L+\varepsilon) - \Phi^{-1}(L)$ lies between 0 and $\varepsilon/\Phi'(\xi) \to 0$ (mean value theorem again, with $\xi \ge x_1/2 \to \infty$ for large y). Adding,

$$\mathcal{T}(y) - \mathrm{ssqrt}(y) \;=\; (x - x_1) + (x_1 - x_0) \;\longrightarrow\; 0 + 1.$$

(iii) From (ii) and $\mathrm{ssqrt}(y) = \mathrm{e}^{W(L)} = L/W(L)$ it suffices to expand W at large argument: the fixed-point relation $W = \ln L - \ln W$ iterated twice from $W_0 = \ln L$ gives $W(L) = \ln L - \ln\ln L + \frac{\ln\ln L}{\ln L}\big(1 + O\big(\frac{\ln\ln L}{\ln L}\big)\big)$, whence

$$\frac{L}{W(L)} \;=\; \frac{L}{\ln L}\Big(1 + \frac{\ln\ln L}{\ln L} + O\Big(\big(\tfrac{\ln\ln L}{\ln L}\big)^2\Big)\Big).$$

Since $\mathcal{T} = \mathrm{ssqrt} + 1 + o(1)$ and the correction $1 + o(1)$ is negligible against $\frac{L}{\ln L}\cdot\frac{\ln\ln L}{\ln L} \to \infty$, (32) follows with $L = \ln y$, $\ln L = \ln\ln y$, $\ln\ln L = \ln\ln\ln y$. □

Tables 3 and 4 display the two regimes; Figure 6 shows the right one on a log-log scale, where $\mathcal{T}$ runs parallel to $\mathrm{ssqrt}(y) \sim \ln y/W(\ln y)$ and visibly above $\ln y$. A sharper probe of (31): since the shift satisfies (from the proof) $x_1 - x_0 \approx W/(W+1)$ with $W = W(\ln y)$, the excess $\mathcal{T} - \mathrm{ssqrt}$ should track $W(\ln y)/(W(\ln y)+1)$; at $y = 10^{12}, 10^{24}, 10^{48}$ the pairs read (0.7199, 0.7085), (0.7521, 0.7459), (0.7792, 0.7759): the two columns close on each other and on 1, but only at logarithmic speed, because $W(\ln y)$ itself grows like $\ln\ln y$. At the googol level, $\mathcal{T}(10^{100}) = 57.7645882431$, while the two-term formula (32) predicts 55.51: iterated logarithms converge so slowly that even a googol leaves the third term contributing 2.3.

Table 3: Left asymptotics (30): the measured $\ln\bigl(\mathcal{T}(y)/\mathrm{e}^{y}\bigr)$ against the prediction $-\frac{y^2}{2}\mathrm{e}^{y}$. By $y=-20$ the agreement is to five significant digits.

y	$\ln\bigl(\mathcal{T}(y)/\mathrm{e}^{y}\bigr)$	$-\frac{y^2}{2}\mathrm{e}^{y}$
-5	-0.079444	-0.084224
-10	-0.0022655	-0.0022700
-20	-4.12231×10^{-7}	-4.12231×10^{-7}
-40	-3.39868×10^{-15}	-3.39868×10^{-15}

Table 4: Right asymptotics: $\mathcal{T}(y)$ and the excess over the super square root, tending to 1 (plus-one law). The entry $T(10^9)\approx 10$ is explained by $\mathcal{T}\bigl((10^{10}-1)/10\bigr)=10$ exactly: the argument 999,999,999.9 is nearly 10^9. The convergence in the last column is famously slow: $\mathcal{T}-\mathrm{ssqrt}$ reaches only 0.74 at 10^{18}, consistent with the predicted slowness of $W(\ln y)\to\infty$.

y	$\mathcal{T}(y)$	$\mathcal{T}(y)-\mathrm{ssqrt}(y)$
10^3	5.1933138	0.6377781
10^6	7.7478016	0.6820049
10^9	10.0000000	0.7049131
10^{12}	12.0872259	0.7199081
10^{18}	15.9616322	0.7393752

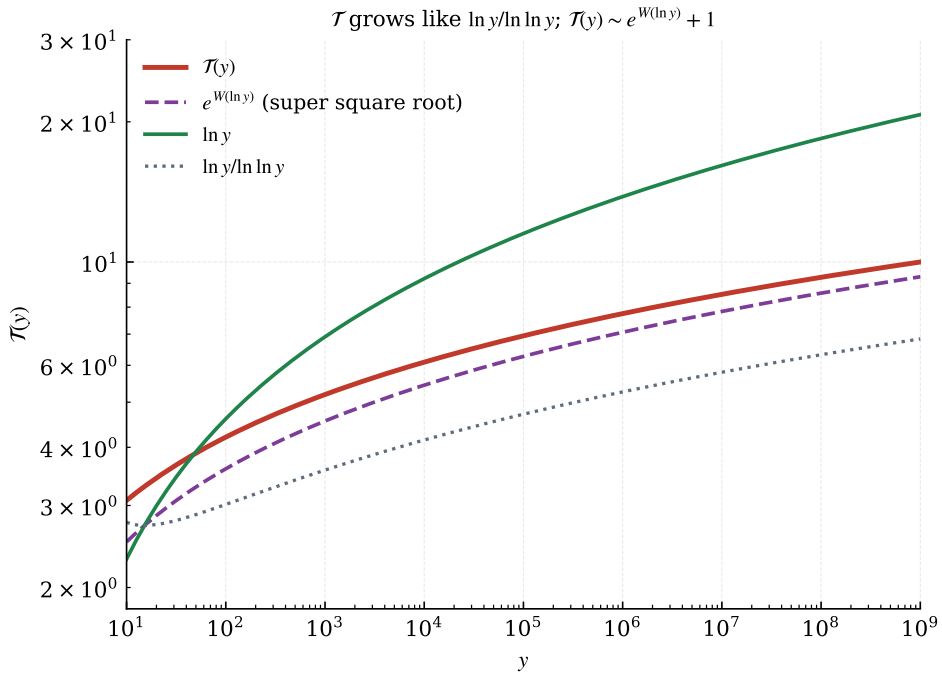


Figure 6: Growth of $\mathcal{T}$ at large arguments (log-log scale): $\mathcal{T}(y)$ (solid red) runs just above the square super root $\mathrm{e}^{W(\ln y)}$ (dashed purple), in accordance with the plus-one law (31), and far below any power of y: its order is $\ln y/\ln\ln y$ (dotted curves).

11 Curiosities: integrals, a table of values, numerology, and open problems

11.1 Integrals

Areas under $\mathcal{T}$ reduce, by the substitution $y = f(x)$, to areas under x^{x-1}.

Proposition 11.1.

$$\int_0^1 \mathcal{T}(y)\,\mathrm{d}y \;=\; \alpha + \ln\alpha - \int_1^\alpha x^{x-1}\,\mathrm{d}x \;=\; 1.432359924510\ldots, \tag{33}$$

$$\int_{-1}^0 \bigl(1-\mathcal{T}(y)\bigr)\,\mathrm{d}y \;=\; 1-\beta+\ln\beta+\int_\beta^1 x^{x-1}\,\mathrm{d}x \;=\; 0.409721130782\ldots \tag{34}$$

In (34) *the integrand at the lower limit takes the value* $\beta^{\beta-1} = (1-\beta)/\beta = v$, *the constant of Theorem 7.5.*

Proof. Substitute $y = f(x)$, $\mathrm{d}y = f'(x)\,\mathrm{d}x$. For (33): $\int_0^1 \mathcal{T}(y)\,\mathrm{d}y = \int_1^\alpha x\,f'(x)\,\mathrm{d}x = \bigl[x f(x)\bigr]_1^\alpha - \int_1^\alpha f(x)\,\mathrm{d}x = \alpha - \int_1^\alpha \bigl(x^{x-1} - \tfrac{1}{x}\bigr)\,\mathrm{d}x$. For (34): $\int_{-1}^0 \bigl(1-\mathcal{T}(y)\bigr)\,\mathrm{d}y = \int_\beta^1 (1-x)f'(x)\,\mathrm{d}x = \bigl[(1-x)f(x)\bigr]_\beta^1 + \int_\beta^1 f(x)\,\mathrm{d}x = (1-\beta) + \int_\beta^1 x^{x-1}\,\mathrm{d}x + \ln\beta$. The decimal values were computed by high-precision quadrature and re-verified through the right-hand sides. □

Neither $\int x^{x-1}$ nor any relative of it has a known closed form; the classical neighbors are the two *sophomore's dream* integrals [16], $\int_0^1 x^{-x}\,\mathrm{d}x = \sum n^{n-1}/n!$ and $\int_0^1 x^x\,\mathrm{d}x = \sum (-1)^{n+1} n^{-n}$, whose series involve the very powers n^{-n} that populate our parameters b_n. Whether (33) admits an evaluation in terms of known constants is Open Problem (7).

11.2 A table of values

For reference and numerical spot-checks we record Table 5, computed to eighty significant digits and truncated here; the exact entries are from Proposition 7.1 and Theorem 4.2, and every transcendental entry was verified through its defining equation.

11.3 Numerology, debunked

Constants as charismatic as $1.776775\ldots$ attract numerological claims the way Tahawus attracted titanium prospectors. The near-coincidence $\alpha \approx \sqrt{\pi}$ (they share two decimal digits) dissolves under measurement:

$$\alpha - \sqrt{\pi} \;=\; 4.321189\ldots \times 10^{-3},$$

a gap four thousand times wider than the precision to which α is known. A subtler trap, since it survives five digits, is

$$\alpha + \frac{1}{\alpha} - \bigl(\sqrt{\pi} + W(1)\bigr) \;=\; -4.6497 \times 10^{-6},$$

and no doubt with a little effort one could manufacture coincidences surviving eight. The theorem-level response is that α is transcendental (Stanley; Corollary 4.3), so *no* identity expressing α through algebraic numbers can hold exactly; identities mixing α with other transcendental constants are not excluded by the theorem but are ruled out numerically for every natural candidate we tested ($\alpha\beta = 0.5395339\ldots$, $\alpha + \beta = 2.0804341\ldots$, $\alpha - \varphi = 0.1587410\ldots$, $1/\alpha = 0.5628174\ldots$, none of them recognizable). The honest summary: the Tahawus constant is a genuine constant, not a photograph of $\sqrt{\pi}$ wearing a different hat.

Table 5: Values of $\mathcal{T}$ at selected arguments. The entries at $b \in \{0, \frac{3}{2}\}$ are exact (Proposition 7.1); $\mathcal{T}(1) = \alpha$ and $\mathcal{T}(-1) = \beta$ are the constants of the constellation.

b	$\mathcal{T}(b)$	remark
-10	0.0000452971906610539	$\approx \mathrm{e}^{-10}$
-5	0.00622336542629749	
-2	0.105721516538092434	
$-3/2$	0.175410254005848625	
-1	0.303659127029966051	β, A226568, transcendental
$-1/2$	0.556573703488657722	
0	1	exact; $= \varphi_0$
$1/2$	1.45595925301034942	
1	1.77677504009705470	α, A124930, transcendental
$3/2$	2	exact; $= \varphi_{3/2}$
2	2.16575937062328429	partner of the root $x = -1$
e	2.34459646238796334	
π	2.42877946150189671	
5	2.69536745976022816	
10	3.07706224655857979	
100	4.20694896031877337	
10^6	7.74780160879762861	

11.4 Open problems

(1) **No closed form.** We conjecture that no finite expression built from the Lambert W function—or from the generalized Lambert functions of [9]—by composition, arithmetic operations, and inversion solves the super-quadratic $x^x = bx + 1$. The structural obstruction outlined in Section 2 (the unknown occurring as base, exponent, and linearly) is not covered by any existing no-closed-form theorem that we know of.

(2) **Complex roots.** How many complex solutions does $z^z = z + 1$ have (per branch of the power)? By Remark 4.5 all of them are transcendental, but even the count is open; nothing seems to be known about their distribution.

(3) **The radius, rigorously.** Prove that the conjugate critical values $y_c, \overline{y_c}$ of Subsection 6.3 are the nearest singularities of the principal branch of $\mathcal{T}$, making $R = 0.6526100\ldots$ a theorem. The numerical evidence is strong (two independent methods, broad critical-point sweep), but a proof requires control of all critical values of f in the disk of radius R.

(4) **The Maclaurin coefficients.** The coefficients d_n of (16) are rational ($d_3 = -\frac{1}{2}$, $d_6 = -\frac{49}{120}$, $d_{12} = \frac{62041117}{19958400}, \ldots$). Is there a closed form, a hypergeometric-type characterization, or a generating function for them? Is $d_2 = 0$ the only vanishing coefficient? Do the numerators and denominators appear in the OEIS?

(5) **The super-golden family.** For $n \in \mathbb{N}$ let τ_n be the unique root of $x^x = x^n + 1$ in $(n, n+1)$; then $\tau_1 = \alpha$ and $\tau_2 = \delta$, the fixed point of Theorem 8.1. The Gelfond–Schneider step of Theorem 4.2 applies verbatim to every τ_n; what is missing is a uniform exclusion of rational roots. Numerically $\tau_n = n + \big(n^n \ln n\big)^{-1}\big(1 + o(1)\big)$ (already accurate to 4% at $n = 3$): the family approaches the integers super-exponentially fast. Is τ_n transcendental for every n? We have not found $\delta = \tau_2$ or any other τ_n, $n \geq 2$, recorded in the OEIS.

(6) **Complex dynamics of the decremented tower.** Describe the Fatou/Julia dichotomy for $z \mapsto \alpha^z - 1$ on the complex plane: the attracting fixed point 0, the repelling fixed point α (Theorem 9.1), and the escaping set. For which bases $a \in (1, \mathrm{e})$ does the boundary between the basin of 0 and the escaping set disconnect the plane?

(7) **The integral.** Evaluate $\int_1^{\alpha} x^{x-1}\,\mathrm{d}x$ (equivalently (33)) in terms of named constants or series, or prove that no such expression exists in a reasonable class. The same question for the moments $\int_0^1 \mathcal{T}(y)^k\,\mathrm{d}y$.

The Tahawus function began life as a blog post's act of friendship toward a beautiful unsolvable equation; we hope to have shown that it repays systematic study. Every constant it touches—α, β, ν, δ—is transcendental, its graph threads between e^y and $1+y$ with third-order degeneracy at the origin, and its inverse-functional structure ties the Lambert W function, the sophomore's dream, and the golden ratio into one small, coherent constellation. The ghost town has a functioning mathematical namesake at last.

A Deferred elementary arguments

Three small calculus facts were quoted in the body; we prove them here. They share a theme: one-line monotonicity arguments at an interior critical point.

Lemma A.1. *The function $u_1(x) := x\big(1+2\ln x\big)$ satisfies $u_1(x) \le 1$ on $(0,1]$, with equality only at $x=1$.*

Proof. $u_1'(x) = 3 + 2\ln x$ vanishes only at $x_0 = \mathrm{e}^{-3/2}$; u_1 decreases on $(0, x_0)$ and increases on $(x_0, 1]$. Hence on $(0,1]$ its maximum is $\max\{u_1(x_0), u_1(1)\} = \max\{-2\mathrm{e}^{-3/2}, 1\} = 1$, attained only at $x=1$. □

Lemma A.2. *For $s>0$, with $h(s) := (s-1)\mathrm{e}^{-s}$:*

(i) *$h(s) > -1$, and more precisely h increases from -1 (at $s \to 0^+$) to $h(2) = \mathrm{e}^{-2}$ and then decreases back toward 0^+; in particular $h(s) \in (-1, \mathrm{e}^{-2}]$;*

(ii) *$q(s) := (s-1)^2\mathrm{e}^{-s} < 1$ for all $s>0$, with $\sup_{s>0} q(s) = \lim_{s\to 0^+} q(s) = 1$ not attained.*

Proof. (i) $h'(s) = (2-s)\mathrm{e}^{-s}$: positive on $(0,2)$, negative on $(2,\infty)$. So h is strictly increasing from $\lim_{s\to 0^+} h(s) = -1$ to $h(2) = \mathrm{e}^{-2}$, then strictly decreasing with $\lim_{s\to\infty} h(s) = 0$; the range is $(-1, \mathrm{e}^{-2}]$.

(ii) $q'(s) = (s-1)(2-s)\mathrm{e}^{-s}$: q decreases on $(0,1)$, increases on $(1,2)$, decreases on $(2,\infty)$, with $q(0^+) = 1$, $q(1) = 0$, $q(2) = \mathrm{e}^{-2} < 1$, $q(\infty) = 0$. Hence $\sup_{s>0} q = \max\{q(0^+), q(2)\} = 1$, approached only as $s \to 0^+$; for every $s>0$, $q(s) < 1$. □

Lemma A.3. $(\ln x)^2 < x$ *for every* $x \ge 1$.

Proof. Let $k(x) := x - 2\ln x$; then $k'(x) = 1 - 2/x$ vanishes only at $x=2$, and $k(2) = 2 - 2\ln 2 > 0$ (as $\ln 2 < 1$) with $k \to \infty$ at both ends of $(0,\infty)$, so $k>0$, i.e. $2\ln x < x$. Now $g(x) := x - (\ln x)^2$ has $g(1) = 1 > 0$ and $g'(x) = 1 - 2\ln x/x > 0$ by what was just proved, so g is increasing and positive on $[1,\infty)$. □

B Computational notes

All numerical statements in the paper were produced by two Python scripts, `tahawus_research.py` (numbers) and `tahawus_figures.py` (figures), delivered alongside this article; they use `mpmath` [17] at 80 significant digits and `sympy` for the exact symbolic checks. The core evaluator brackets the root of $x^x = yx+1$ by bisection and polishes it with Newton, robust down to the far left tail, where the bisection is performed in $t = \ln x$:

```
def f(x):  return (x**x - 1)/x

def T(y):
    y = mp.mpf(y)
    if y > 0:
        lo, hi = mp.mpf(1), mp.mpf(2)      # root in [1, inf)
        while f(hi) < y: hi *= 2
```

```
    elif y > -5:
        lo, hi = mp.mpf('1e-40'), mp.mpf(1)  # root in (0, 1]
    else:                                    # bisect in t = ln x
        tlo, thi = y - 2, mp.mpf(0)
        g = lambda t: (mp.e**(t*mp.e**t) - 1)*mp.e**(-t) - y
        for _ in range(300):
            tm = (tlo + thi)/2
            if g(tm) < 0: tlo = tm
            else:         thi = tm
        return mp.e**((tlo + thi)/2)
    for _ in range(400):                     # bisection ...
        mid = (lo + hi)/2
        if f(mid) < y: lo = mid
        else:          hi = mid
    return mp.findroot(lambda x: x**x - y*x - 1,   # ... + Newton
                       (lo, hi), tol=mp.mpf(10)**-70)
```

The Maclaurin coefficients of Theorem 6.1 are exact rational numbers computed by the Lagrange–Bürmann recurrence (18) in exact arithmetic (Fraction), with $\varphi(u) = f(1+u)/u$ obtained by exponentiating $E(u) = (1+u)\ln(1+u)$ by convolution, dividing by $1+u$, and reverting:

```
E = [Fr(0)]*(N+1); E[1] = Fr(1)            # (1+u)ln(1+u)
for k in range(2, N+1): E[k] = Fr((-1)**k, k*(k-1))
B = [Fr(0)]*(N+1); B[0] = Fr(1)            # exp(E), by convolutions
for k in range(1, N+1):
    B[k] = sum(i*E[i]*B[k-i] for i in range(1, k+1)) / k
phi = [F[k+1] for k in range(N)]           # phi = f(1+u)/u
inv = [Fr(1)]                              # 1/phi, by reversion
for m in range(1, N):
    inv.append(-sum(phi[i]*inv[m-i] for i in range(1, m+1)))
for n in range(1, N+1):                    # Lagrange-Buermann
    d_n = poly_pow(inv, n, trunc=n-1)[n-1] / n   # phi^{-n} at u^{n-1}
```

The resulting $d_1, \ldots, d_{60}$ are those displayed in (16) (and twenty-four further exact values, used for the radius analysis). The radius determination of Subsection 6.3 combines a least-squares fit of the conjugate-pair recurrence $d_{n+2} = S d_{n+1} - P d_n$ with a direct solve for the critical points:

```
G = lambda x: x**(x+1)*(mp.log(x) + 1 - 1/x) + 1   # = x^2 f'(x)
# (1) fit (S, P) to the 60 exact d_n  ->  singularity y1 = 1/lambda
# (2) continue the inverse branch along arg(y1), then Newton on G:
xc = mp.findroot(G, (seed, seed*1.04))     # conjugate pair of roots
yc = (mp.e**(xc*mp.log(xc)) - 1)/xc        # critical value f(xc)
R  = abs(yc)                               # 0.6526100448447448
```

The six figures are matplotlib vector PDFs (serif fonts, identical evaluator); the scripts are reproduced in the project directory together with the verification harness that re-checks every numerical claim made in the text (OEIS digit strings included). Table 6 collects the constants to sixty digits.

Table 6: Constants of the Tahawus constellation, verified to eighty significant digits (shown truncated); see also Table 5.

constant	value
$\alpha = \mathcal{T}(1)$ (A124930)	1.776775040097054697479730744038756748637411034329296139084374…
$\beta = \mathcal{T}(-1)$ (A226568)	0.303659127029966051245018951213222659991199618761172709644170…
δ (fixed point)	2.228833103278118970618761554677995244189648827644876418 40…
$v = 1/\beta - 1$ (A085846)	2.293166287411861031508028291250805864372257290327121248537710…
$1/\beta$ (A169862)	3.293166287411861031508028291250805864372257290327121248537710…
υ_2 (A144184)	5.507985652773178257589026298052138730016024663304 11…
$\mathcal{T}(2)$	2.165759370623284290556896958558531007686204254108 11803694712…
R (radius of series)	0.652610044844744…
$\mathcal{T}'(\delta)$	0.261543559649…
$g'(\alpha) = \frac{1}{(1+\alpha)(1+\ln\alpha)}$	0.2286830163…
$(1+\alpha)\ln\alpha$ (tower multiplier)	1.5960901426…
$\ln\alpha$ (multiplier at 0)	0.5747999458…

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